

ENM 5310: Data-driven Modeling and Probabilistic Scientific Computing

Gaussian process regression



From linear regression to GPs:

- Linear regression with inputs x_i and outputs y_i : $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$
- Linear regression with M basis functions: $y_i = \sum_{m=1}^M \beta_m \phi_m(x_i) + \epsilon_i$
- Bayesian linear regression with basis functions:

$$\beta_m \sim \mathcal{N}(\cdot | 0, \lambda_m) \quad (\text{independent of } \beta_\ell, \forall \ell \neq m), \quad \epsilon_i \sim \mathcal{N}(\cdot | 0, \sigma^2)$$

- Integrating out the coefficients, β_j , we find:

$$E[y_i] = 0, \quad \text{Cov}(y_i, y_j) = K_{ij} \stackrel{\text{def}}{=} \sum_{m=1}^M \lambda_m \phi_m(x_i) \phi_m(x_j) + \delta_{ij} \sigma^2$$

This is a Gaussian process with covariance function $K(x_i, x_j) = K_{ij}$.

This GP has a finite number (M) of basis functions. Many useful GP kernels correspond to infinitely many basis functions (i.e. infinite-dim feature spaces).

A multilayer perceptron (neural network) with infinitely many hidden units and Gaussian priors on the weights $\rightarrow$ a GP (Neal, 1996)

Gaussian process covariance functions (kernels)

$p(f)$ is a **Gaussian process** if for *any* finite subset $\{x_1, \dots, x_n\} \subset \mathcal{X}$, the marginal distribution over that finite subset $p(\mathbf{f})$ has a multivariate Gaussian distribution.

Gaussian processes (GPs) are parameterized by a **mean function**, $\mu(x)$, and a **covariance function**, or **kernel**, $K(x, x')$.

$$p(f(x), f(x')) = \mathbf{N}(\mu, \Sigma)$$

where

$$\mu = \begin{bmatrix} \mu(x) \\ \mu(x') \end{bmatrix} \quad \Sigma = \begin{bmatrix} K(x, x) & K(x, x') \\ K(x', x) & K(x', x') \end{bmatrix}$$

and similarly for $p(f(x_1), \dots, f(x_n))$ where now μ is an $n \times 1$ vector and Σ is an $n \times n$ matrix.

Using Gaussian processes for nonlinear regression

Imagine observing a data set $\mathcal{D} = \{(\mathbf{x}_i, y_i)_{i=1}^n\} = (\mathbf{X}, \mathbf{y})$.

Model:

$$y_i = f(\mathbf{x}_i) + \epsilon_i$$

$$f \sim \text{GP}(\cdot|0, K)$$

$$\epsilon_i \sim \text{N}(\cdot|0, \sigma^2)$$

Prior on f is a GP, likelihood is Gaussian, therefore posterior on f is also a GP.

We can use this to make predictions

$$p(y_*|\mathbf{x}_*, \mathcal{D}) = \int p(y_*|\mathbf{x}_*, f, \mathcal{D}) p(f|\mathcal{D}) df$$

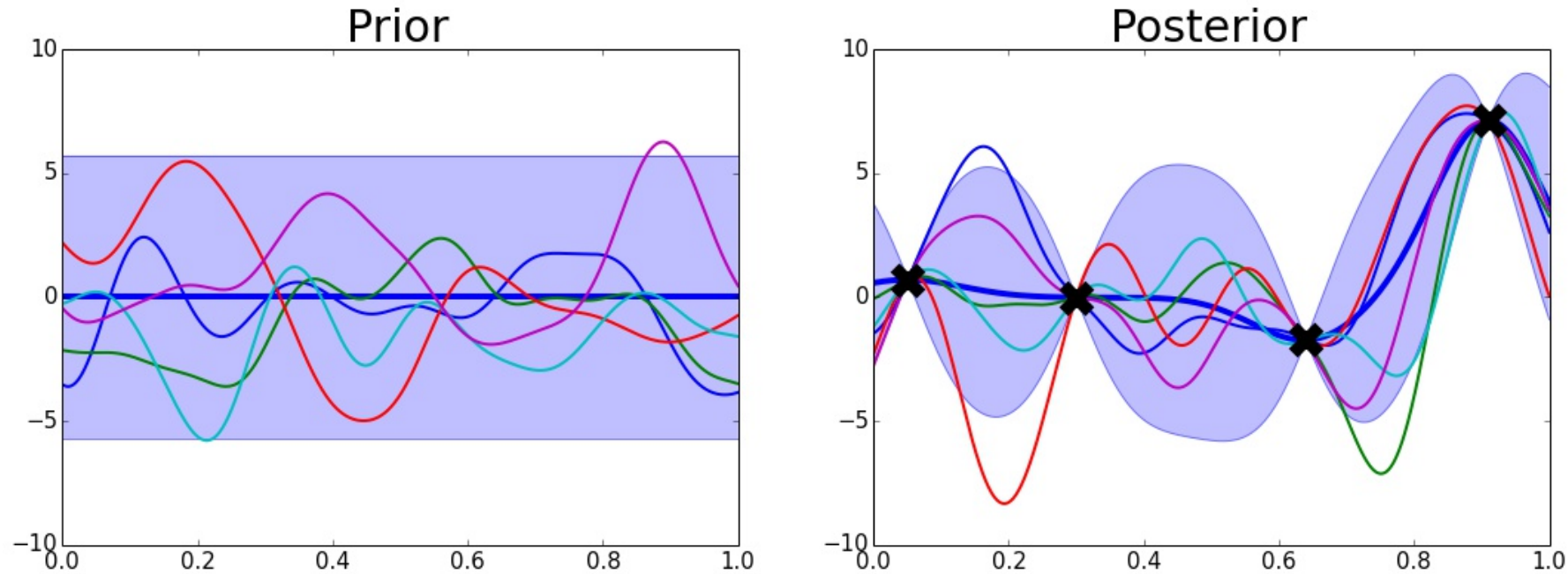
We can also compute the marginal likelihood (evidence) and use this to compare or tune covariance functions

$$p(\mathbf{y}|\mathbf{X}) = \int p(\mathbf{y}|f, \mathbf{X}) p(f) df$$

Data-driven modeling with Gaussian processes

$$y = f(\mathbf{x}) + \epsilon$$

$$f \sim \mathcal{GP}(0, k(\mathbf{x}, \mathbf{x}'; \boldsymbol{\theta}))$$



Training via maximizing the marginal likelihood

$$\log p(\mathbf{y}|\mathbf{X}, \boldsymbol{\theta}) = -\frac{1}{2} \log |\mathbf{K} + \sigma_{\epsilon}^2 \mathbf{I}| - \frac{1}{2} \mathbf{y}^T (\mathbf{K} + \sigma_{\epsilon}^2 \mathbf{I})^{-1} \mathbf{y} - \frac{N}{2} \log 2\pi$$

Prediction via conditioning on available data

$$p(f_* | \mathbf{y}, \mathbf{X}, \mathbf{x}_*) = \mathcal{N}(f_* | \mu_*, \sigma_*^2),$$

$$\mu_*(\mathbf{x}_*) = \mathbf{k}_{*N} (\mathbf{K} + \sigma_{\epsilon}^2 \mathbf{I})^{-1} \mathbf{y},$$

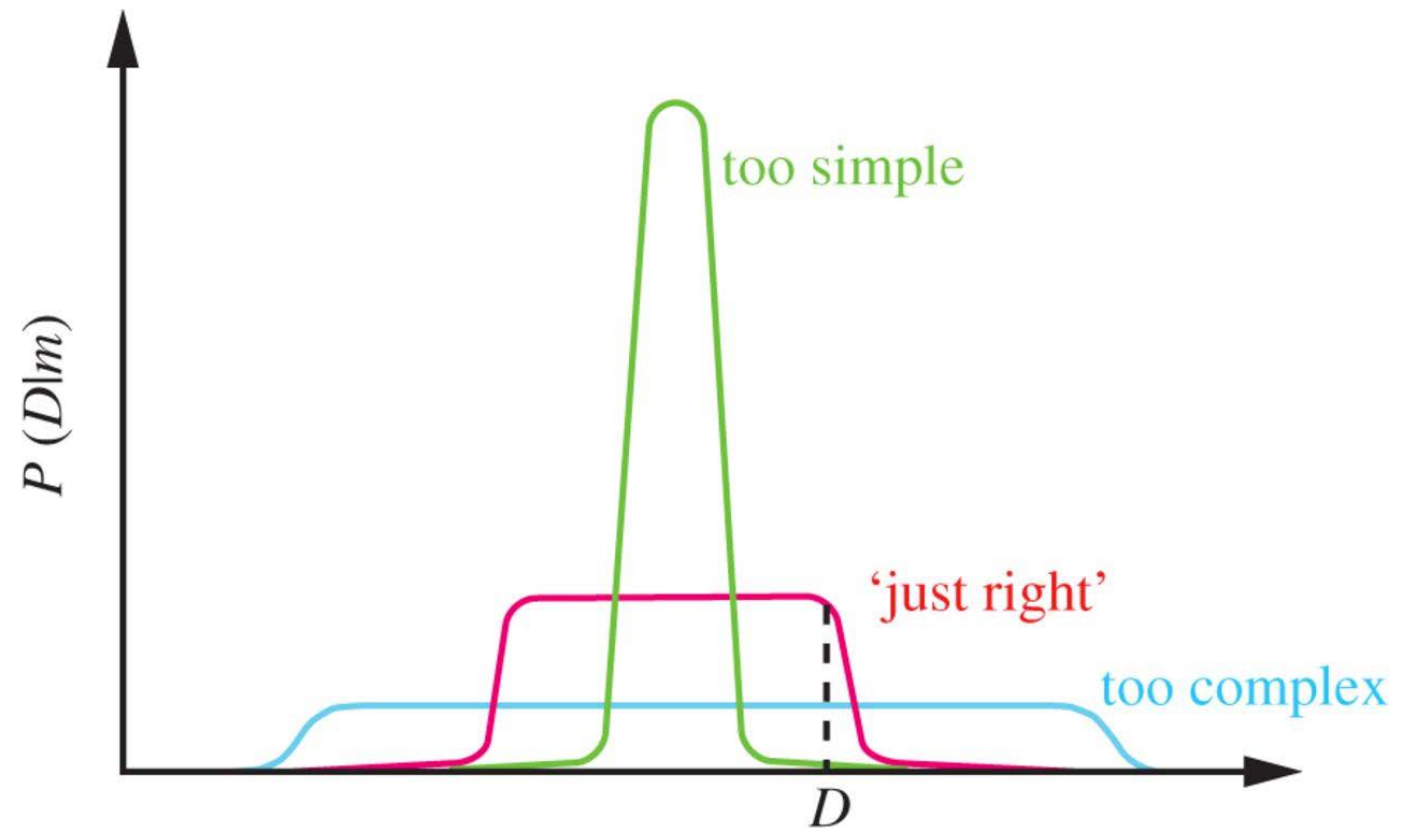
$$\sigma_*^2(\mathbf{x}_*) = \mathbf{k}_{**} - \mathbf{k}_{*N} (\mathbf{K} + \sigma_{\epsilon}^2 \mathbf{I})^{-1} \mathbf{k}_{N*},$$

Occam's razor

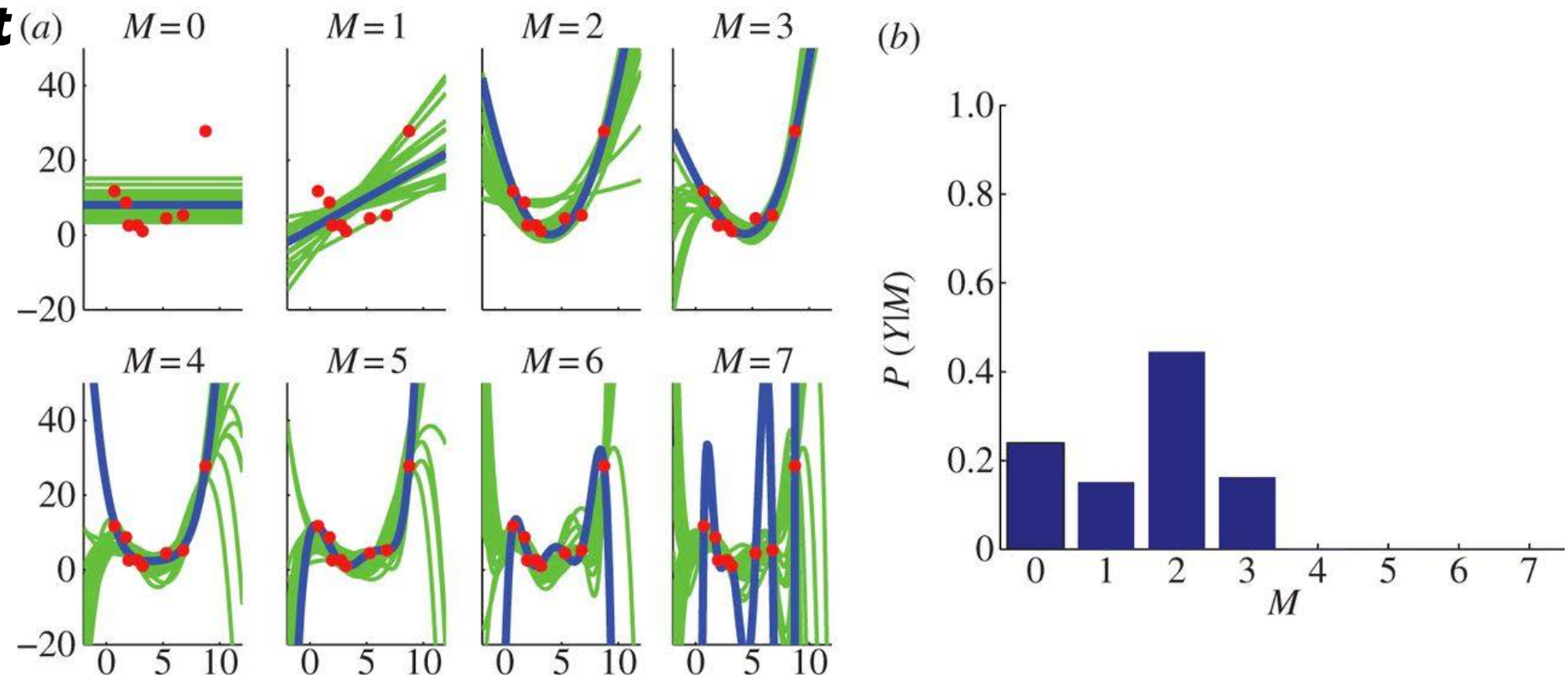
William of Ockham (~1285-1347 A.D)



“plurality should not be posited without necessity.”

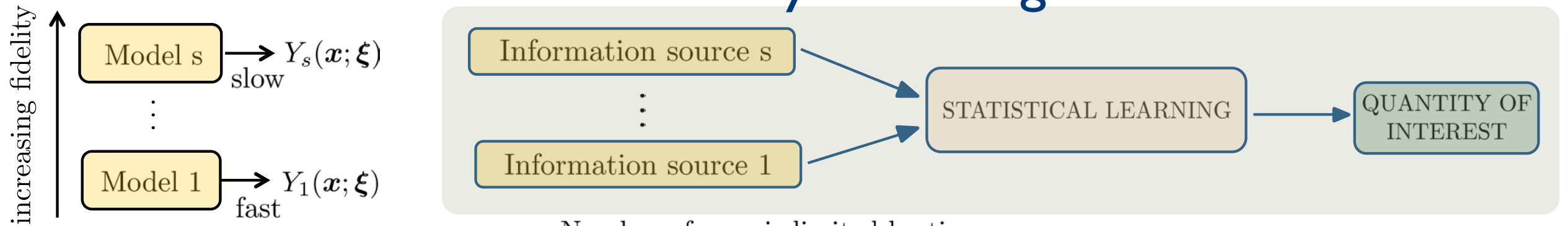


all possible datasets of size n



Ghahramani, Z. (2013). Bayesian non-parametrics and the probabilistic approach to modelling. *Phil. Trans. R. Soc. A*, 371(1984), 20110553.

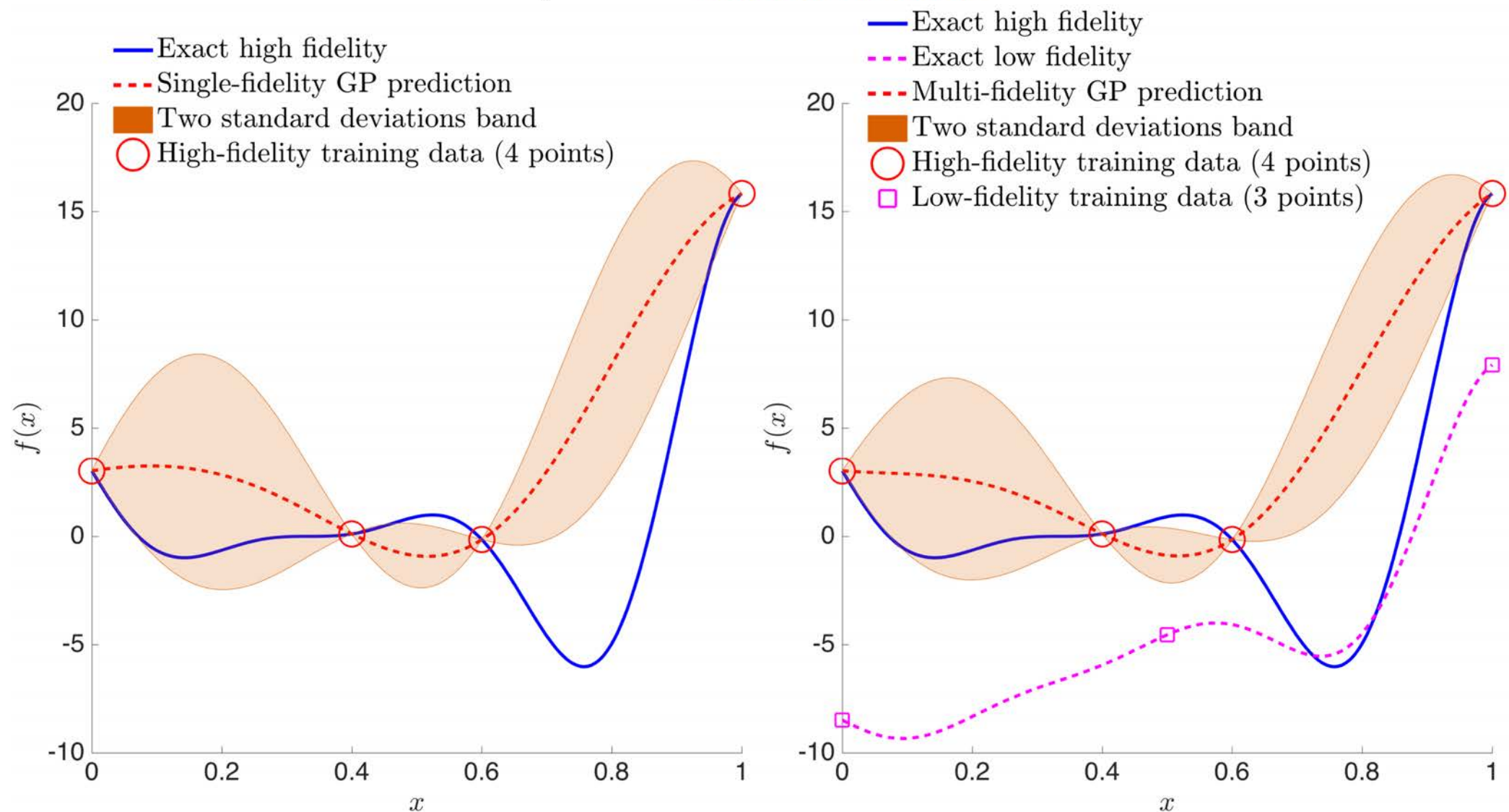
Multi-fidelity modeling



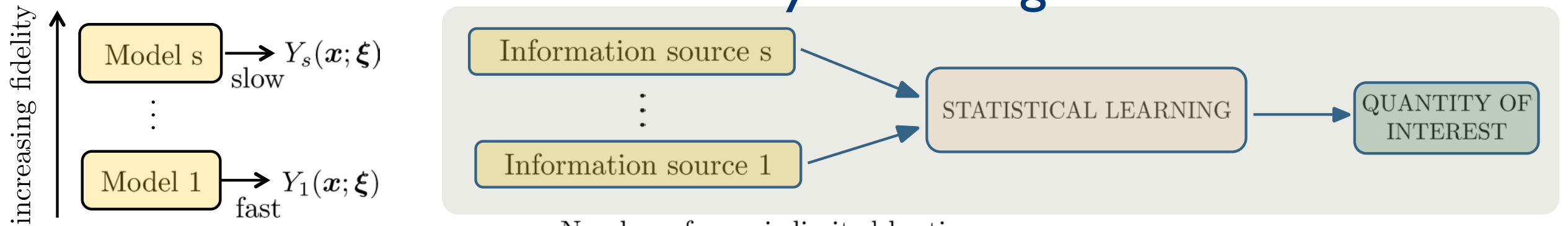
Number of runs is limited by time and computational resources

We cannot compute at all $(x; \xi)$

Prediction of $Z_i(x) = \mathbb{E}[f(Y_i(x; \xi))]$ is a problem of **statistical inference**



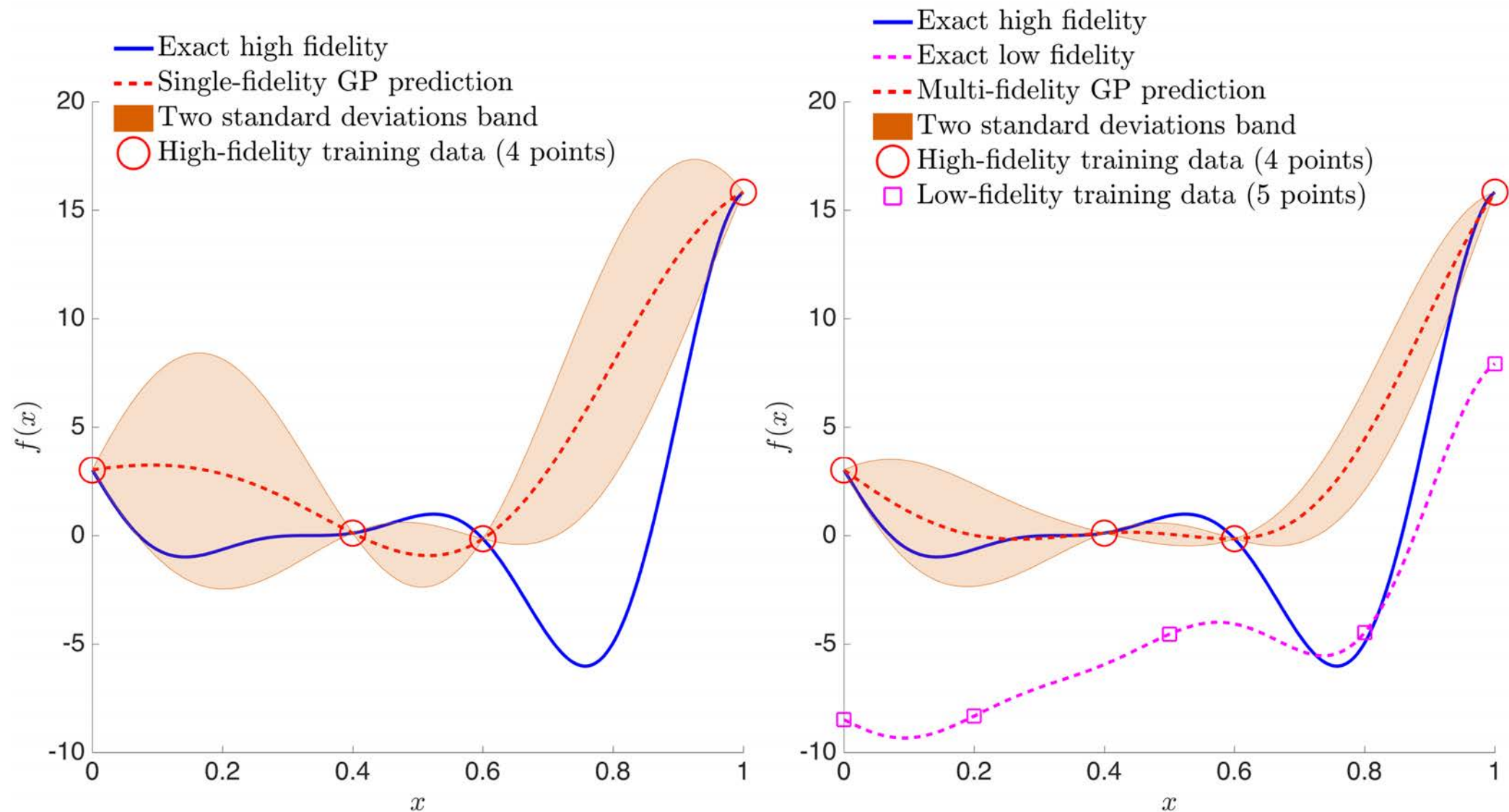
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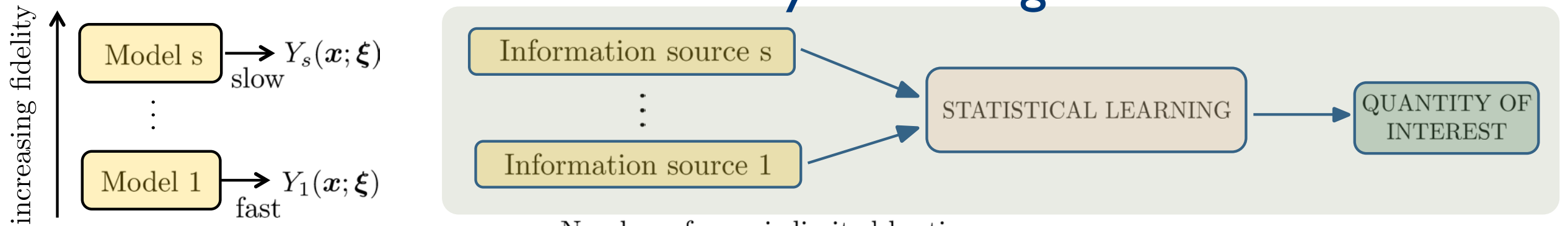
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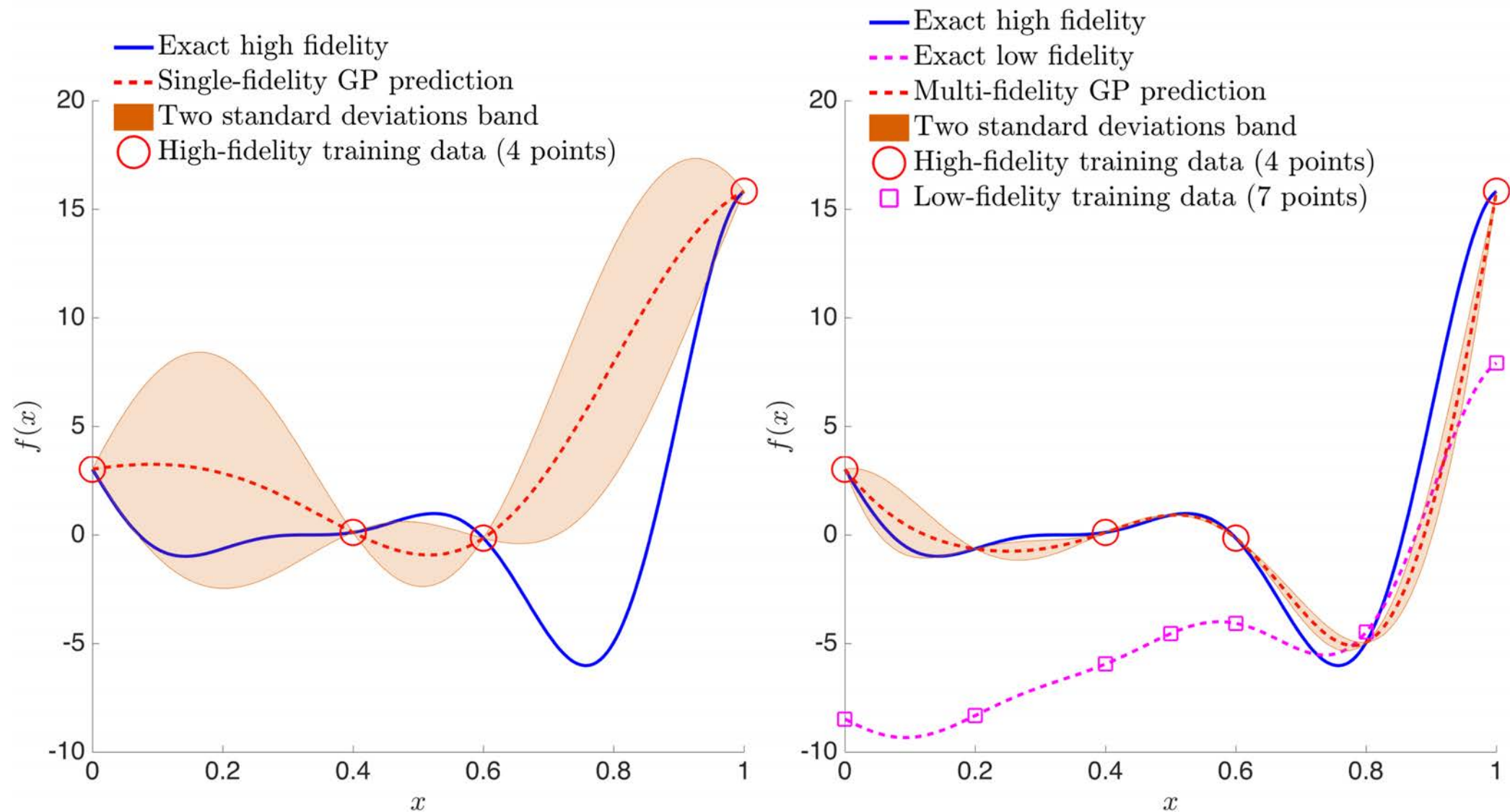
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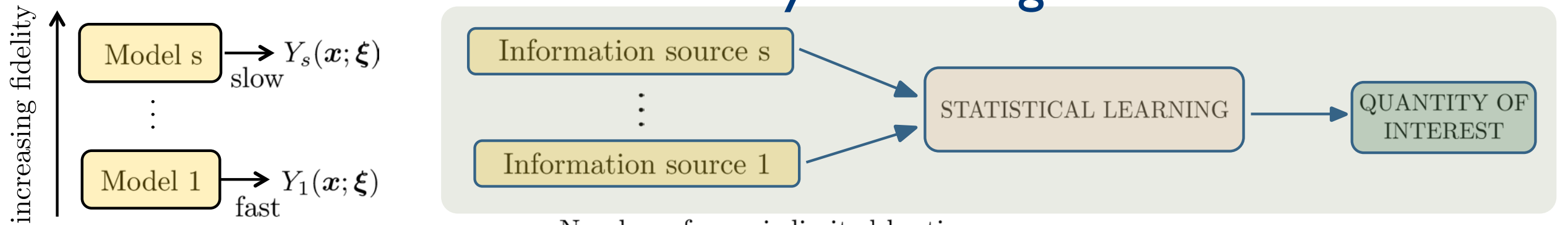
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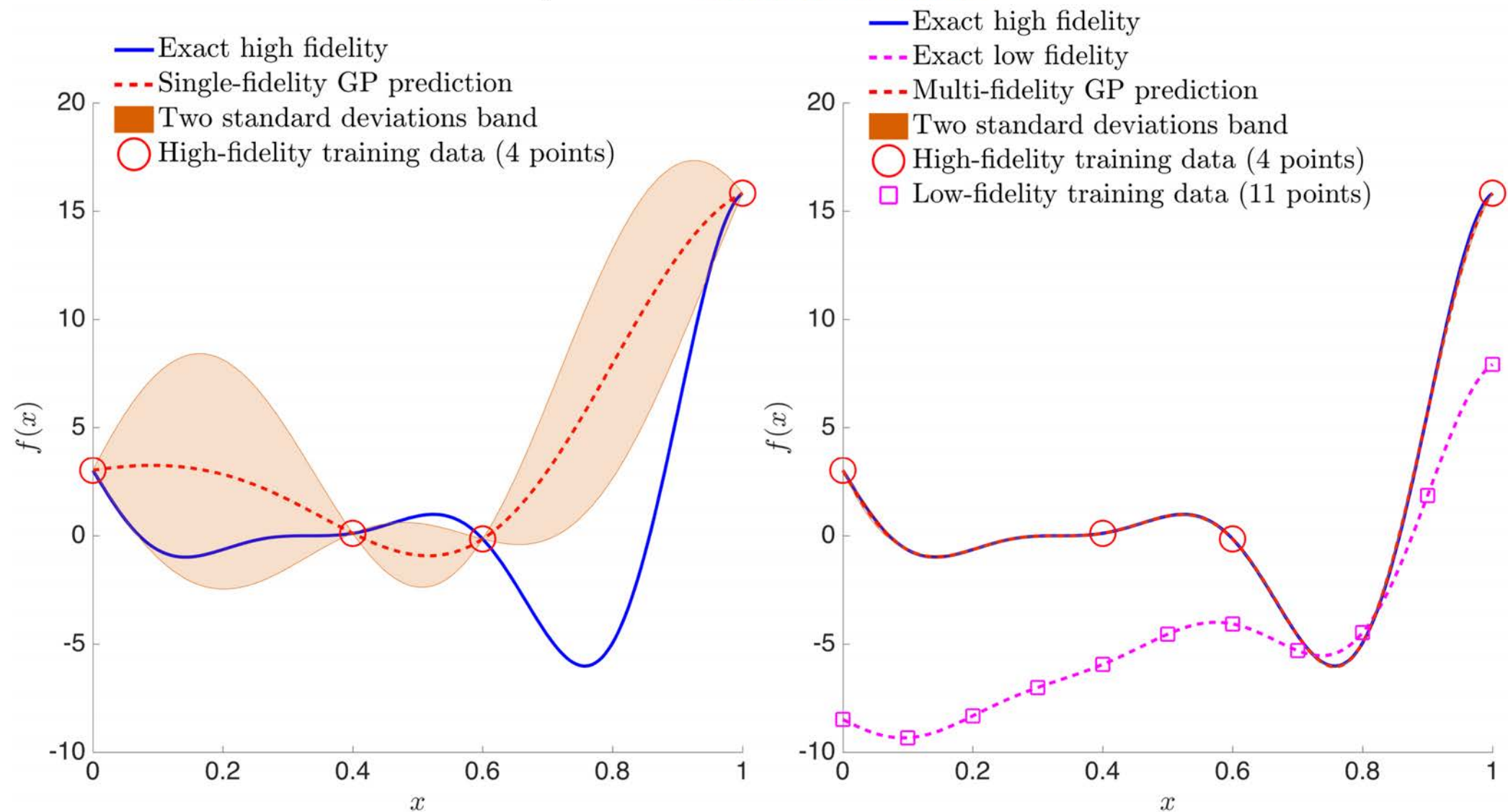
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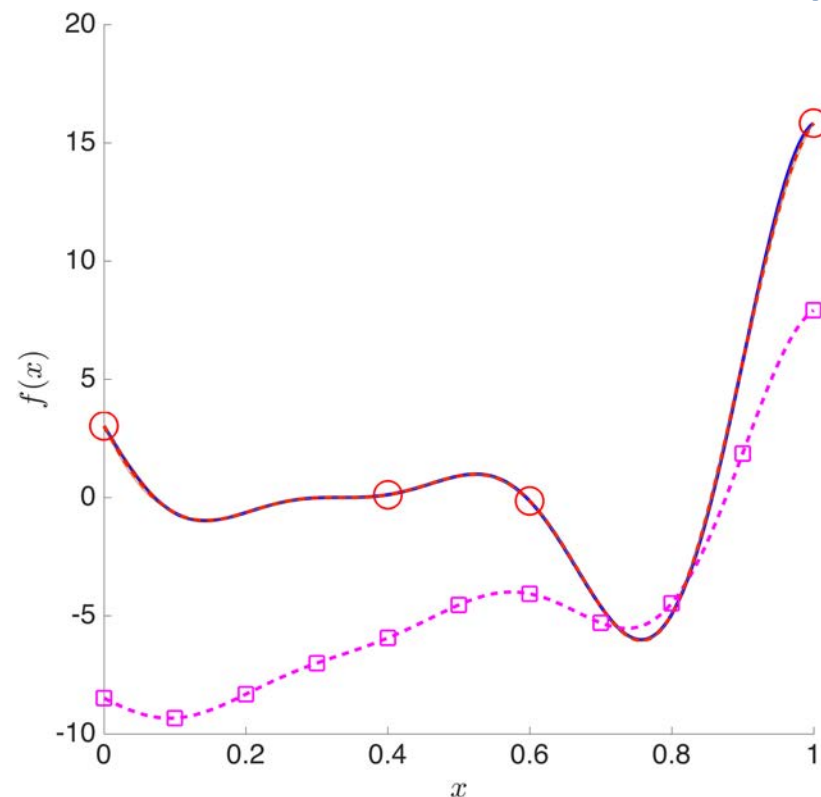
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Multi-fidelity modeling



Multi-fidelity observations:

$$\mathbf{y}_L = f_L(\mathbf{x}_L) + \epsilon_L$$

$$\mathbf{y}_H = f_H(\mathbf{x}_H) + \epsilon_H$$

Probabilistic model:

$$f_H(\mathbf{x}) = \rho f_L(\mathbf{x}) + \delta(\mathbf{x})$$

$$f_L(\mathbf{x}) \sim \mathcal{GP}(0, k_L(\mathbf{x}, \mathbf{x}'; \theta_L))$$

$$\delta(\mathbf{x}) \sim \mathcal{GP}(0, k_H(\mathbf{x}, \mathbf{x}'; \theta_H))$$

$$\epsilon_L \sim \mathcal{N}(0, \sigma_{\epsilon_L}^2 \mathbf{I})$$

$$\epsilon_H \sim \mathcal{N}(0, \sigma_{\epsilon_H}^2 \mathbf{I})$$

Training:

$$\mathbf{y} = \begin{bmatrix} \mathbf{y}_L \\ \mathbf{y}_H \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} \mathbf{0} \\ \mathbf{0} \end{bmatrix}, \begin{bmatrix} k_L(\mathbf{x}_L, \mathbf{x}'_L; \theta_L) + \sigma_{\epsilon_L}^2 \mathbf{I} & \rho k_L(\mathbf{x}_L, \mathbf{x}'_H; \theta_L) \\ \rho k_L(\mathbf{x}_H, \mathbf{x}'_L; \theta_L) & \rho^2 k_L(\mathbf{x}_H, \mathbf{x}'_H; \theta_L) + k_H(\mathbf{x}_H, \mathbf{x}'_H; \theta_H) + \sigma_{\epsilon_H}^2 \mathbf{I} \end{bmatrix} \right)$$

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_L \\ \mathbf{x}_H \end{bmatrix} \quad -\log p(\mathbf{y}|\mathbf{X}, \theta_L, \theta_H, \rho, \sigma_{\epsilon_L}^2, \sigma_{\epsilon_H}^2) = \frac{1}{2} \log |\mathbf{K}| + \frac{1}{2} \mathbf{y}^T \mathbf{K}^{-1} \mathbf{y} - \frac{N_L + N_H}{2} \log 2\pi$$

Prediction:

$$p(f(\mathbf{x}^*)|\mathbf{y}, \mathbf{X}, \mathbf{x}^*) \sim \mathcal{N}(f(\mathbf{x}^*)|\mu(\mathbf{x}^*), \sigma^2(\mathbf{x}^*))$$

$$\mu(\mathbf{x}^*) = \mathbf{k}(\mathbf{x}^*, \mathbf{X}) \mathbf{K}^{-1} \mathbf{y}$$

$$\sigma(\mathbf{x}^*) = \mathbf{k}(\mathbf{x}^*, \mathbf{x}^*) - \mathbf{k}(\mathbf{x}^*, \mathbf{X}) \mathbf{K}^{-1} \mathbf{k}(\mathbf{X}, \mathbf{x}^*)$$

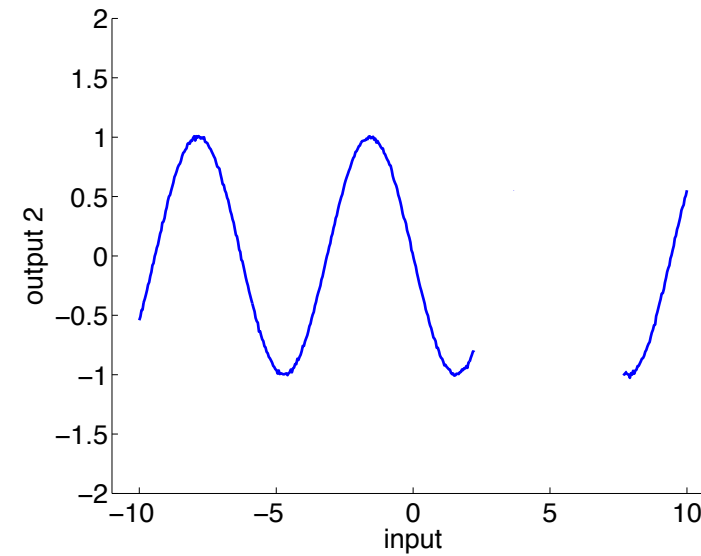
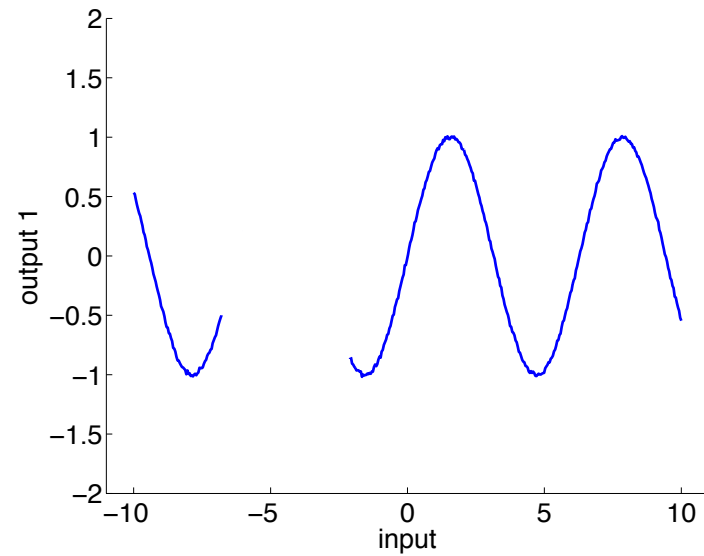
M.C Kennedy, and A. O'Hagan. *Predicting the output from a complex computer code when fast approximations are available*, 2000.

Demo code: <https://github.com/PredictiveIntelligenceLab/GPTutorial>

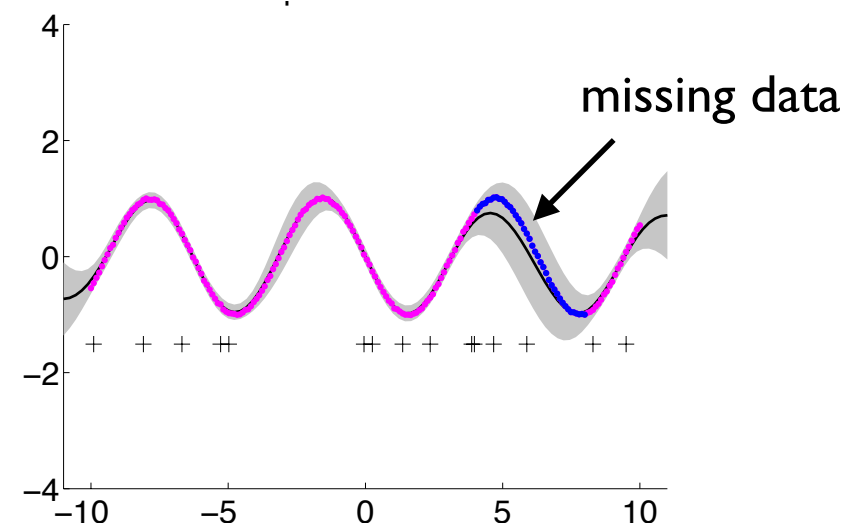
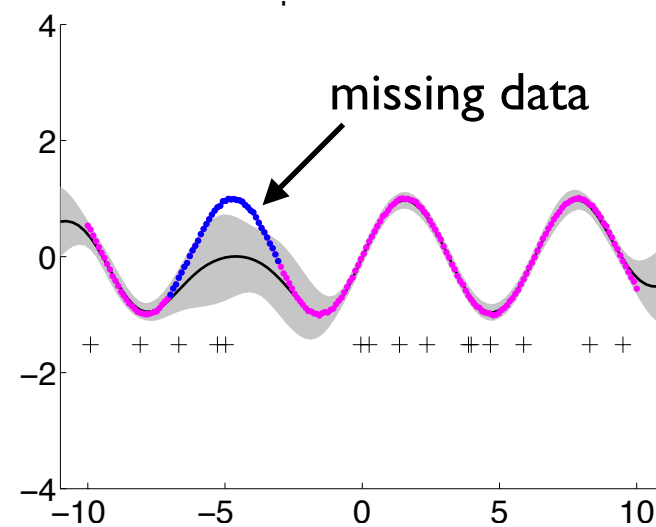
Multi-output Gaussian process regression

Learn two correlated tasks (outputs) with lots of data + missing data

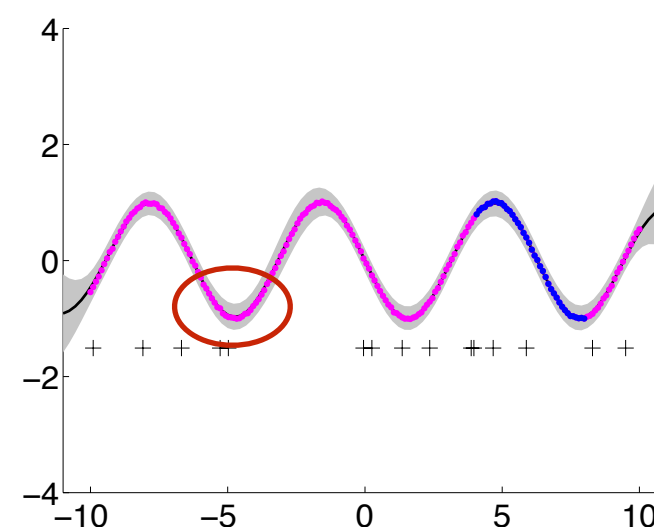
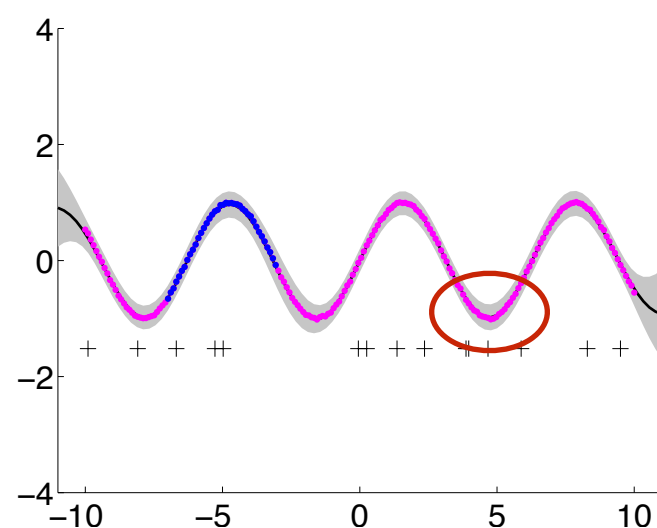
$$\text{output 2} = -\text{output 1}$$



independent
learning

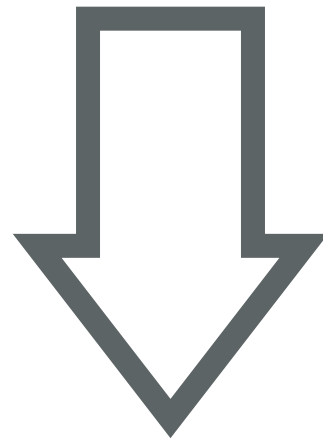


COGP



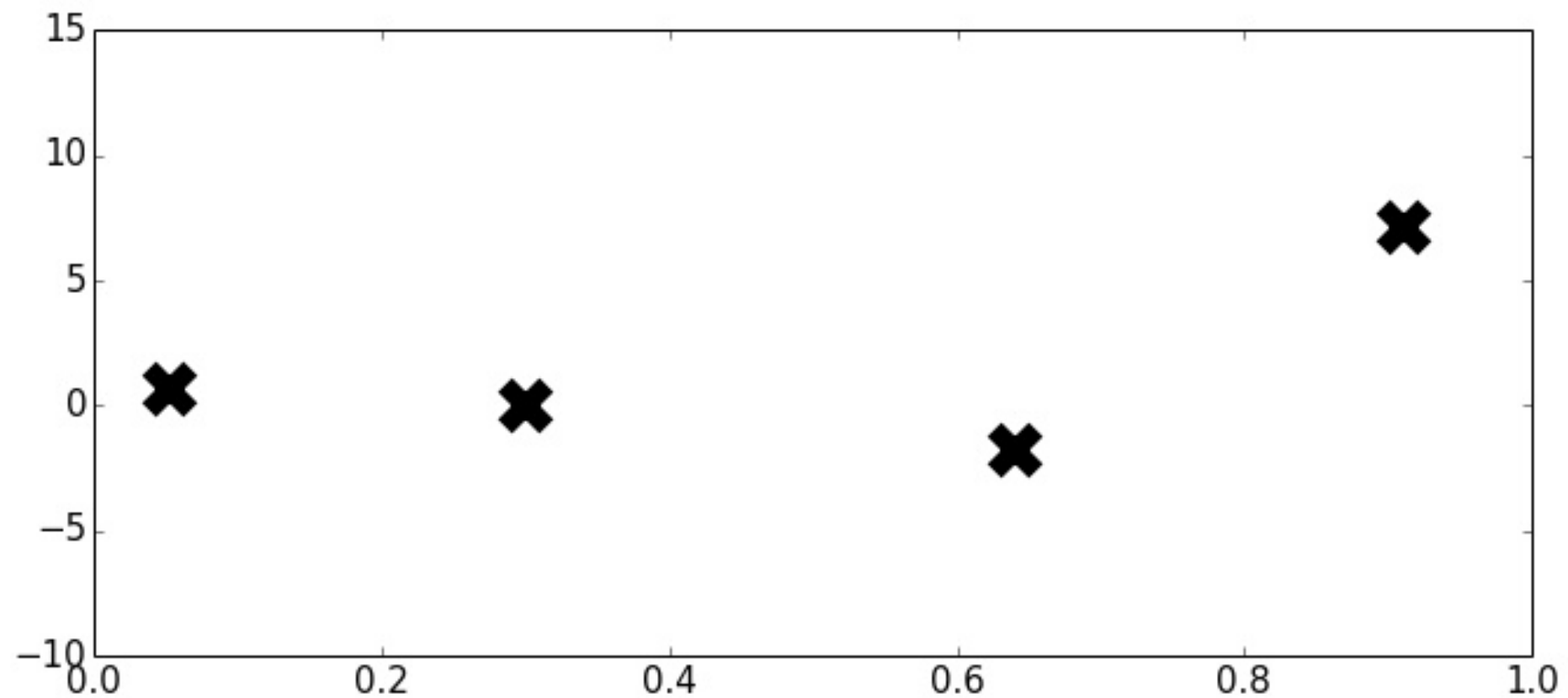
Linear constraints

$$f \sim \mathcal{GP}(0, k(x, x'; \theta))$$



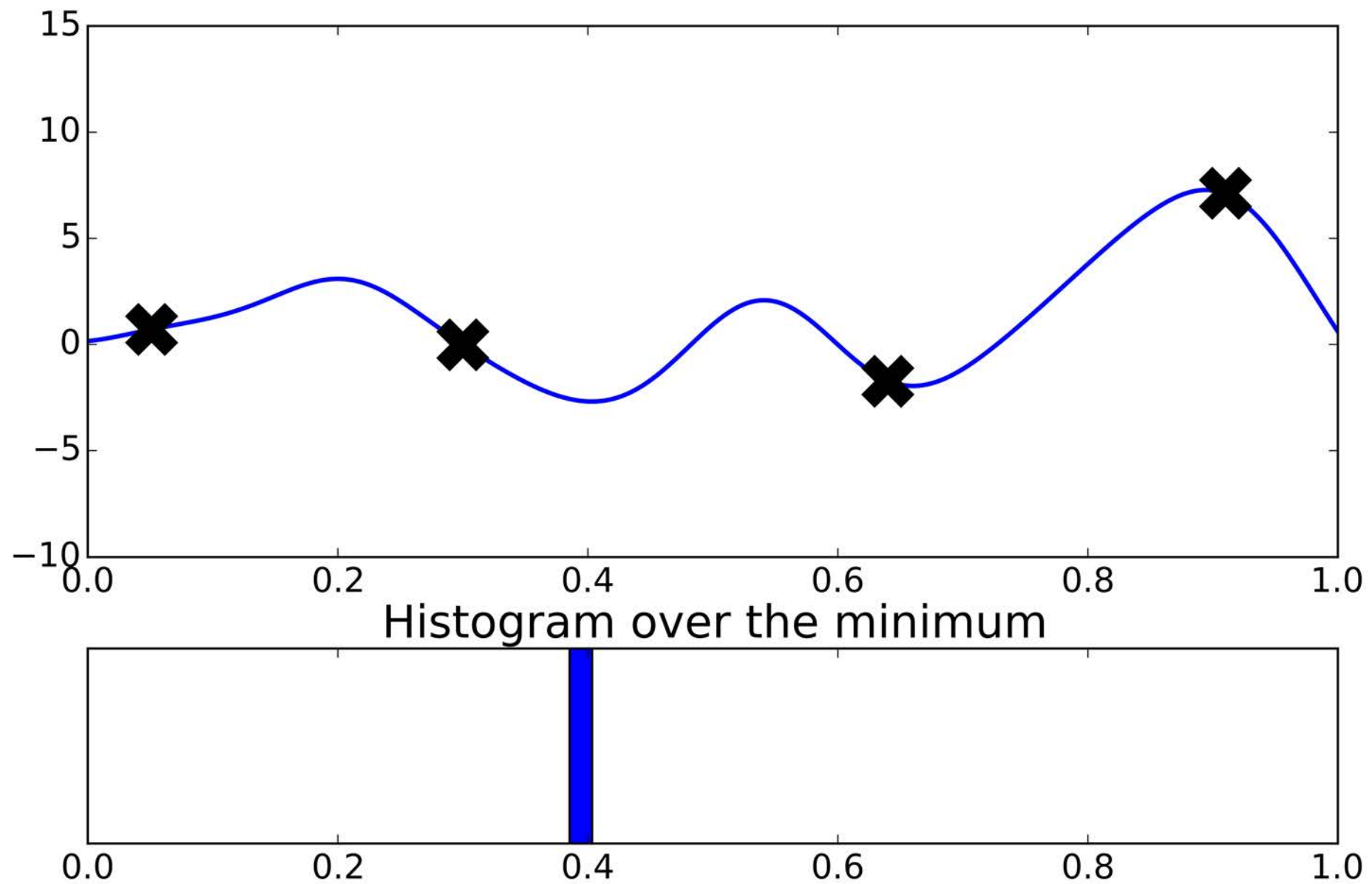
$$\mathcal{L}f \sim \mathcal{GP}(0, \mathcal{L}_x \mathcal{L}_{x'} k(x, x'; \theta))$$

Bayesian optimization

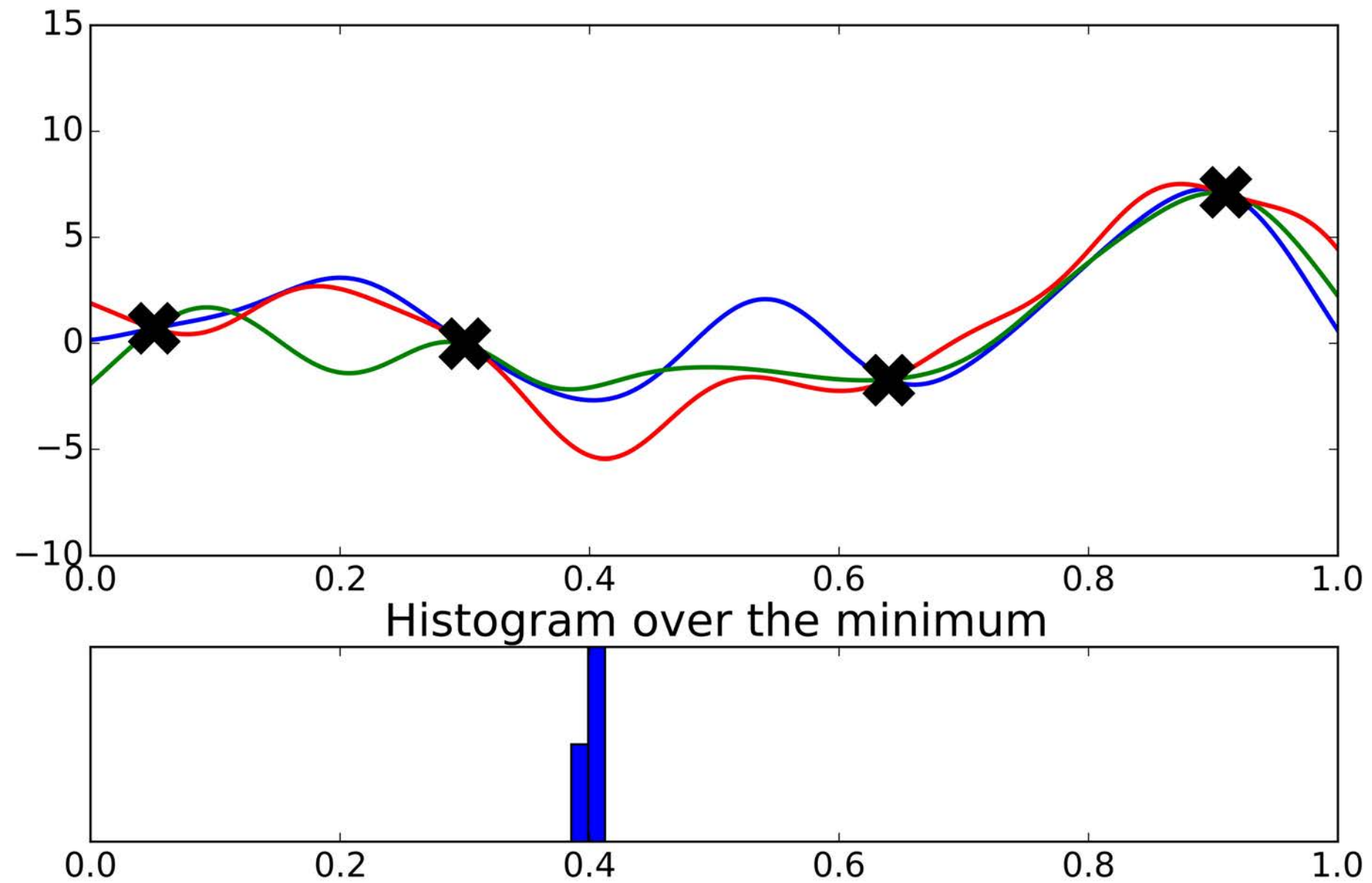


Where is the minimum of f ?
Where should the take the next evaluation?

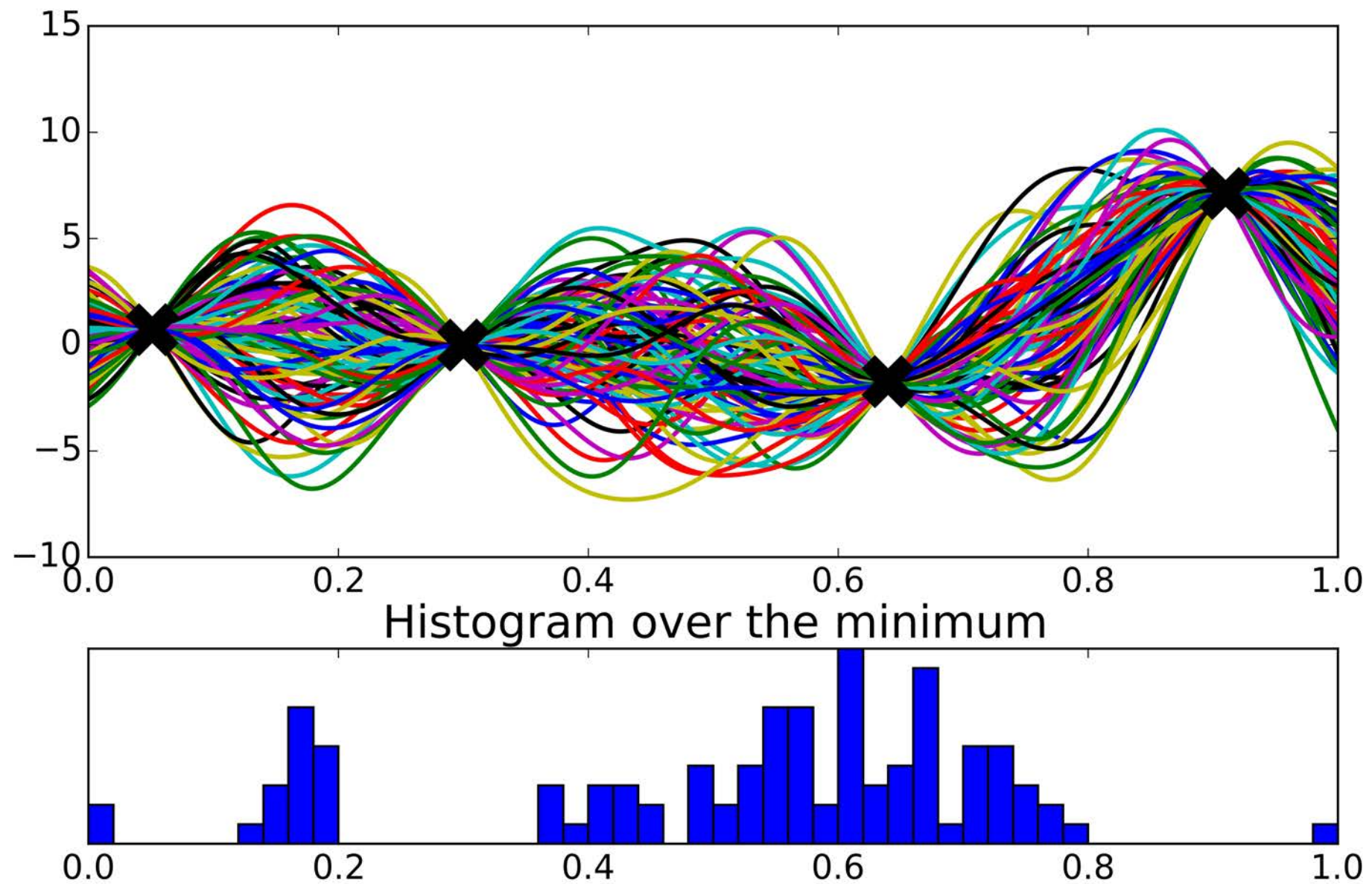
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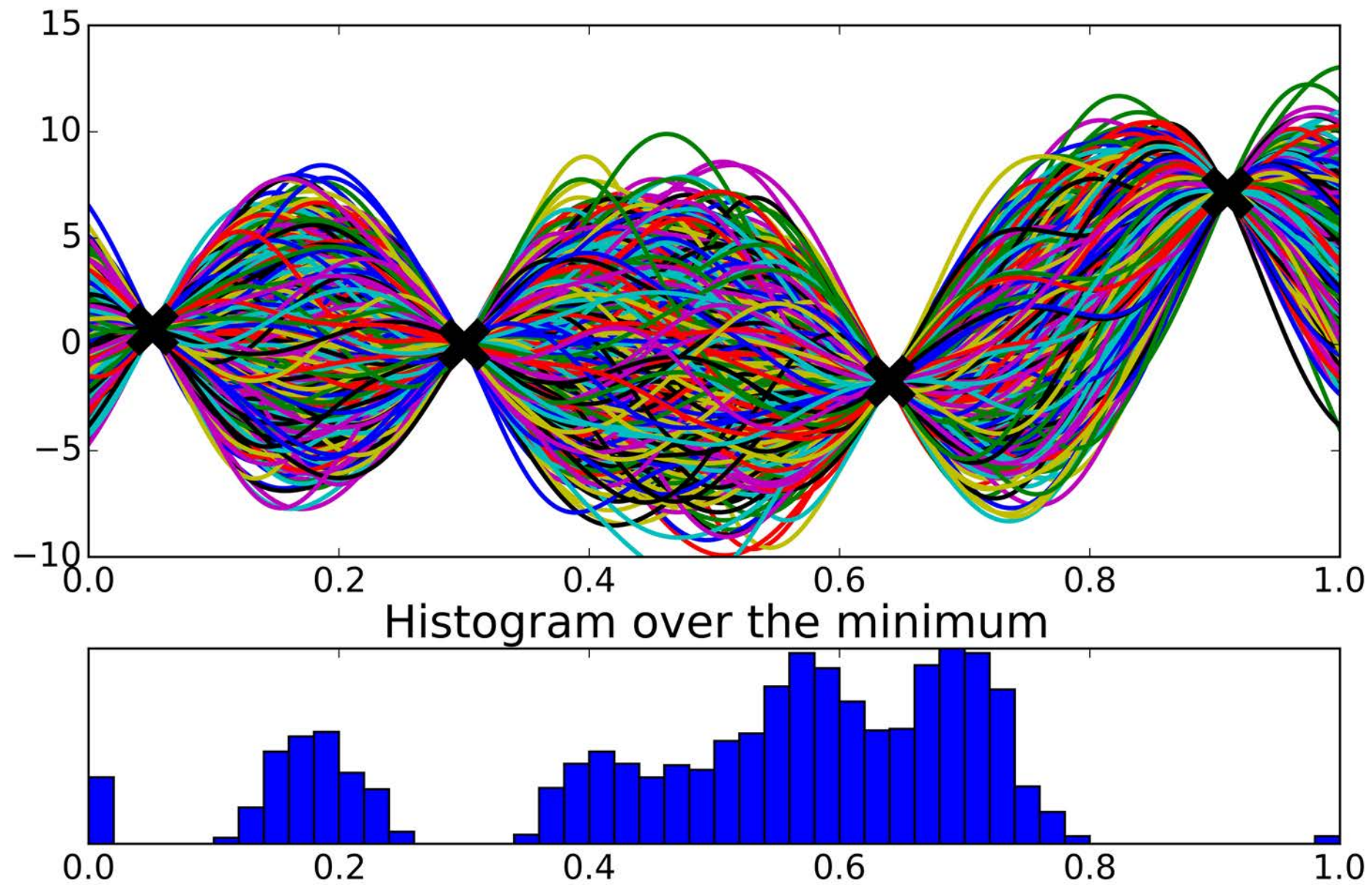
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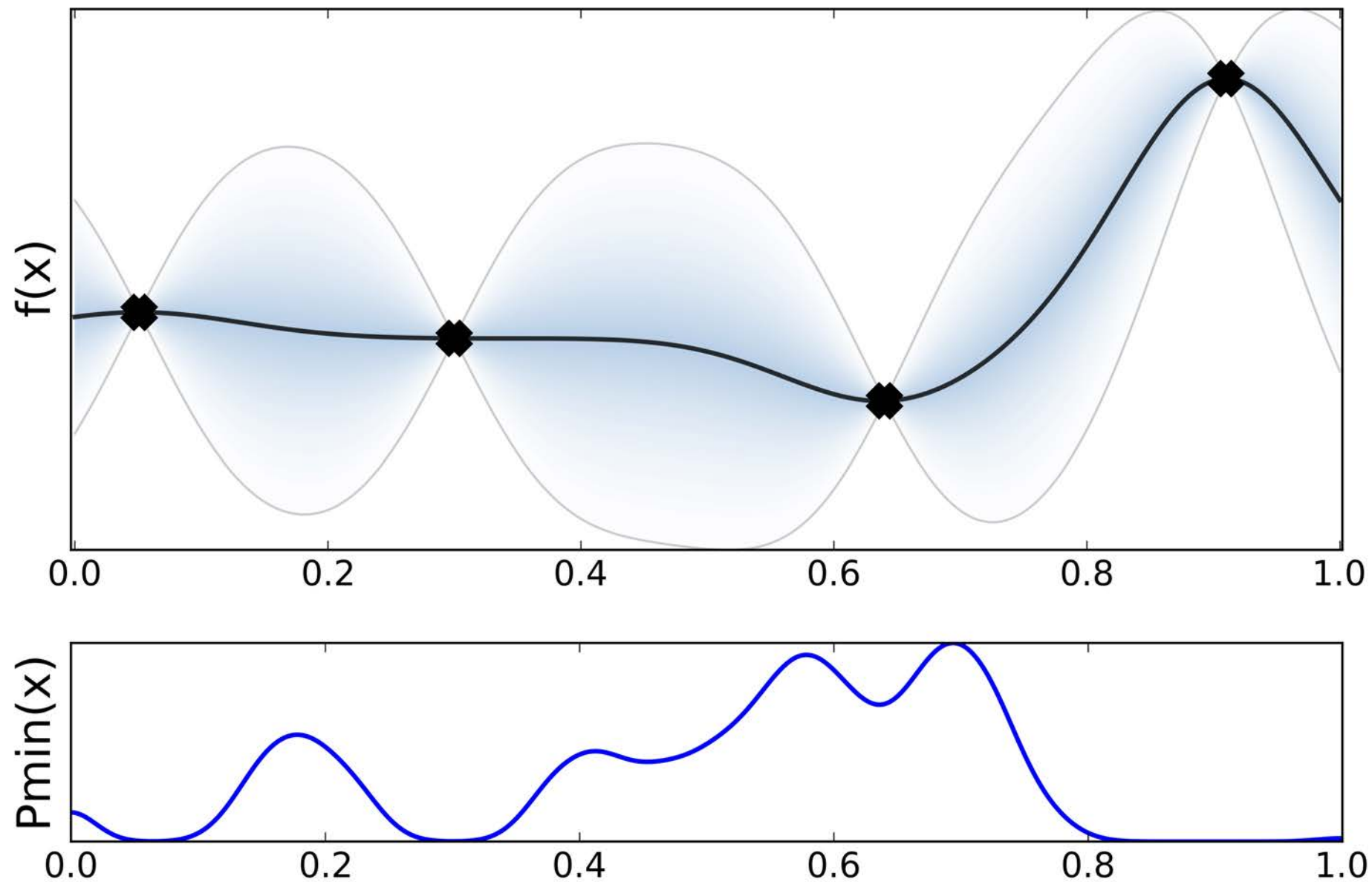
Bayesian optimization



Bayesian optimization



Bayesian optimization



$x^* = \arg \min f(x)$..but $p(x^* | \mathcal{D})$ is not tractable.

Bayesian optimization

Goal: Estimate the global minimum of a function: $\mathbf{x}^* = \arg \min_{\mathbf{x} \in \mathbb{R}^d} g(\mathbf{x})$ (potentially intractable)

Setup: $g(\mathbf{x})$ is a black-box and expensive to evaluate objective function, noisy observations, no gradients.

Idea: Approximate $g(\mathbf{x})$ using a GP surrogate: $y = f(\mathbf{x}) + \epsilon$, $f \sim \mathcal{GP}(f|0, k(\mathbf{x}, \mathbf{x}'; \theta))$

Utilize the posterior to guide a sequential or parallel sampling policy by optimizing a chosen expected utility function

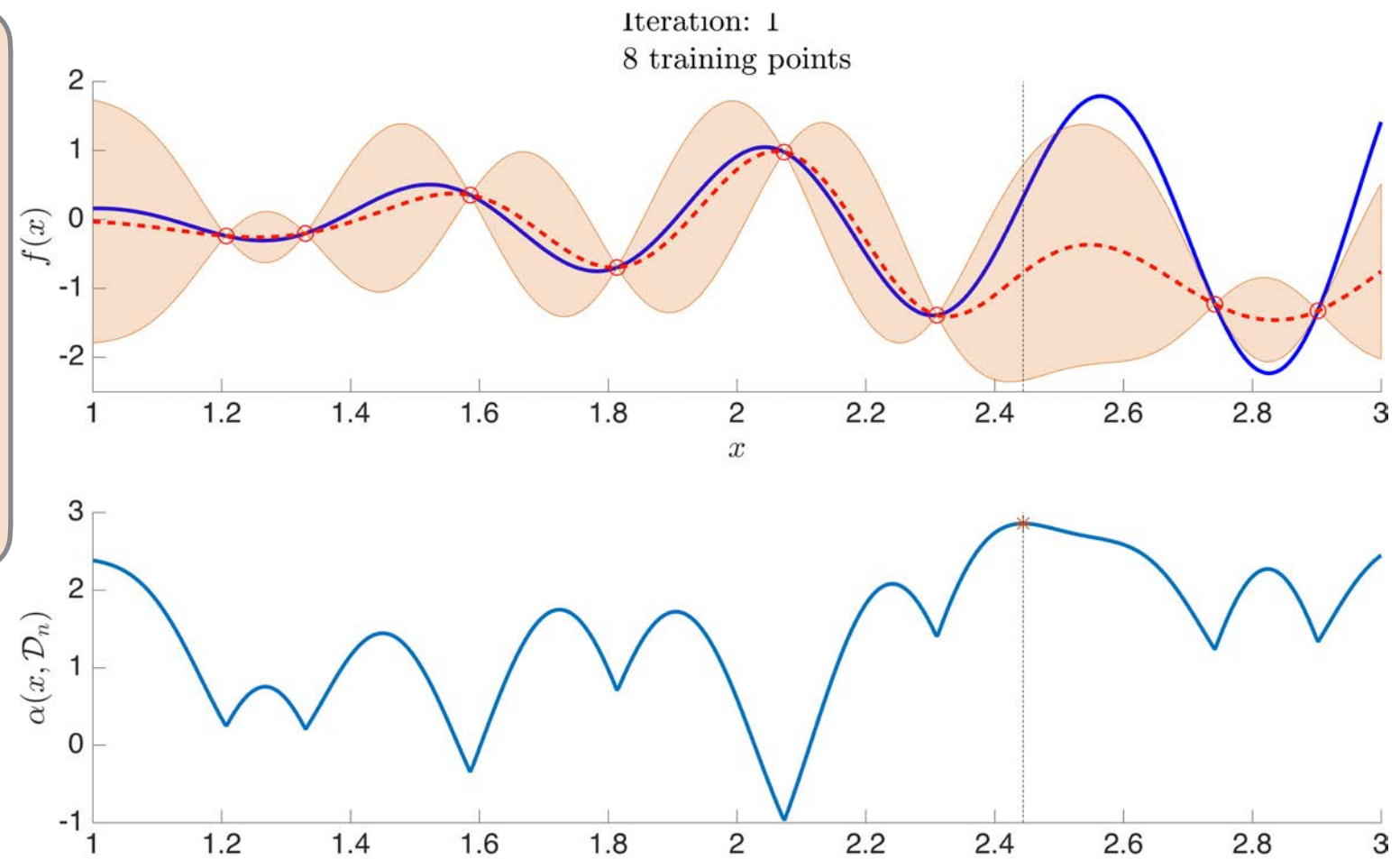
$$\alpha(\mathbf{x}; \mathcal{D}_n) = \mathbb{E}_{\theta} \mathbb{E}_{v | \mathbf{x}, \theta} [U(\mathbf{x}, v, \theta)]$$

The optimization problem is transformed to:

$$\mathbf{x}_{n+1} = \arg \max_{\mathbf{x}} \alpha(\mathbf{x}; \mathcal{D}_n)$$

Remark:

Acquisition functions aim to balance the trade-off between exploration and exploitation.



e.g. sample at the locations that minimize the lower super-quintile risk confidence bound

$$\mathbf{x}_{n+1} = \arg \min_{\mathbf{x} \in \mathbb{R}^d} \mu(\mathbf{x}) - \frac{\phi(\Phi^{-1}(\alpha))}{1 - \alpha} \sigma(\mathbf{x})$$

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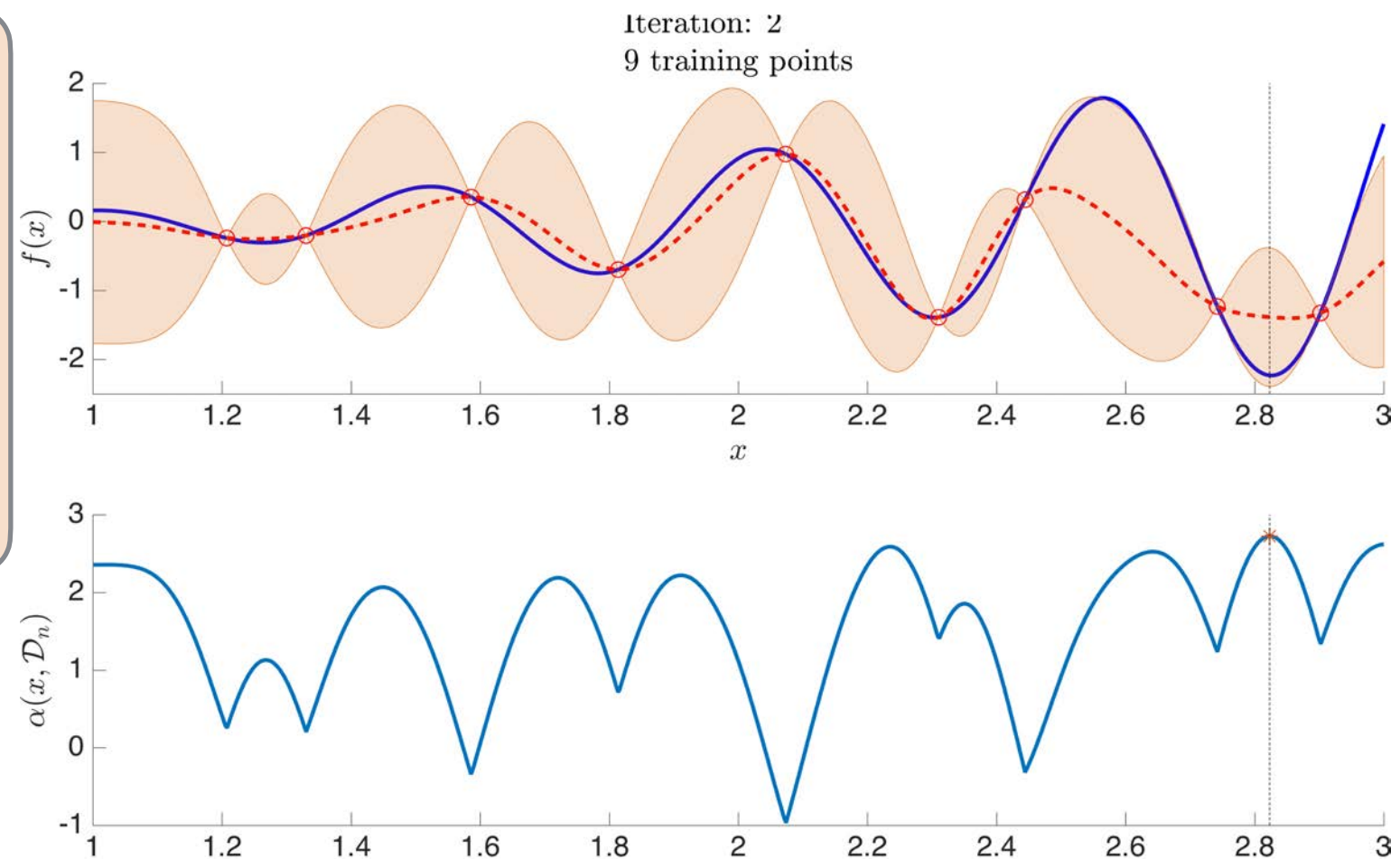
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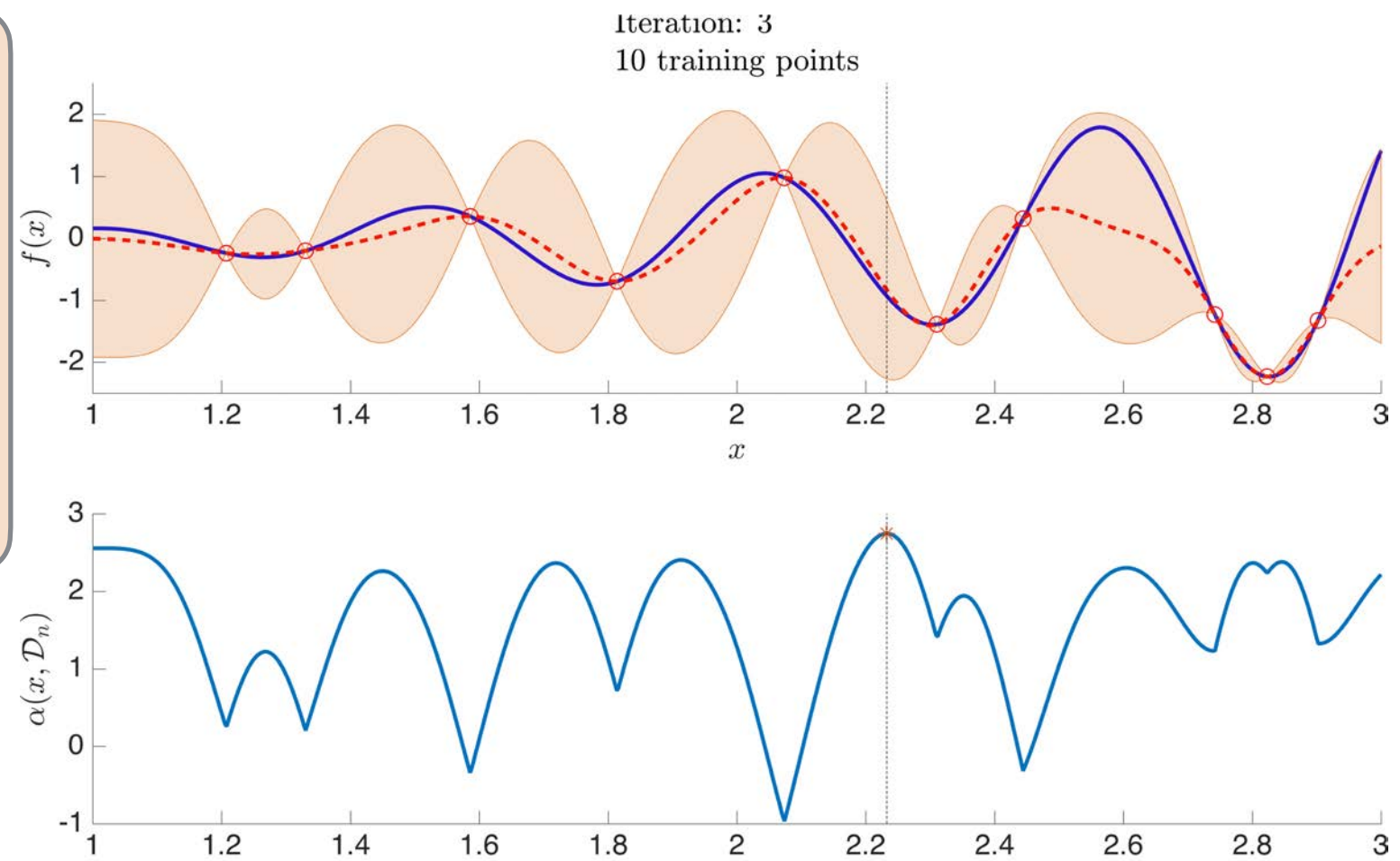
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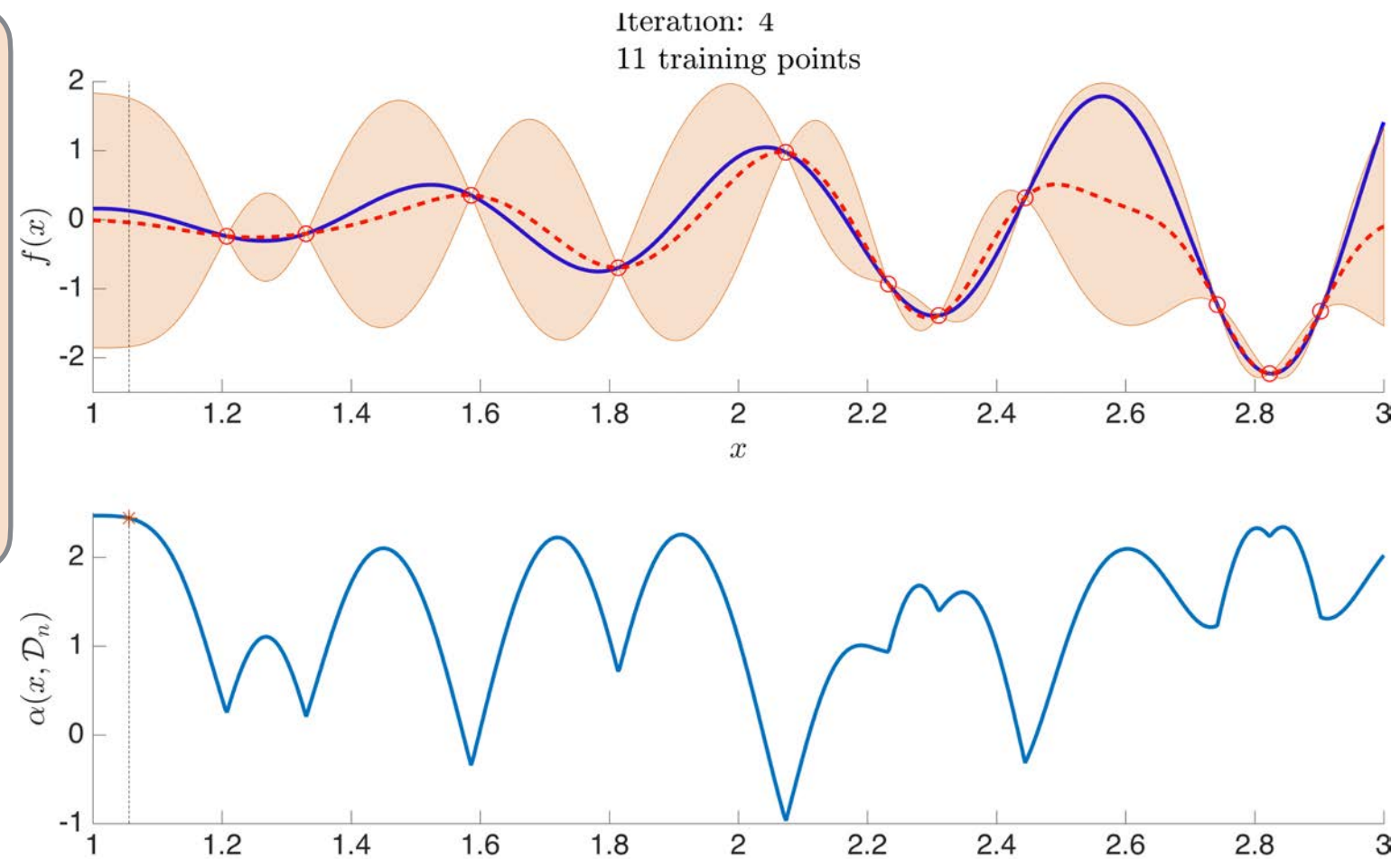
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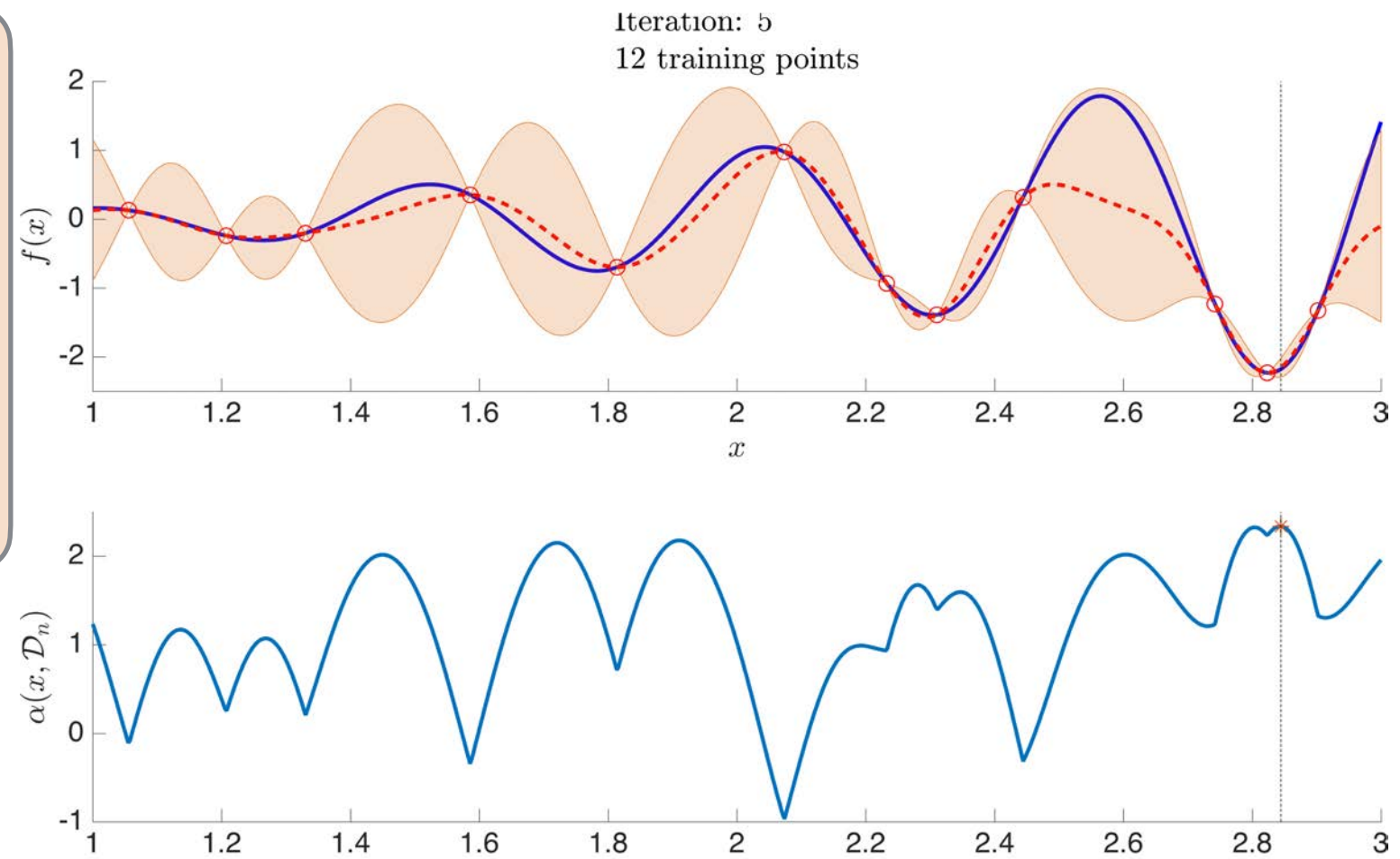
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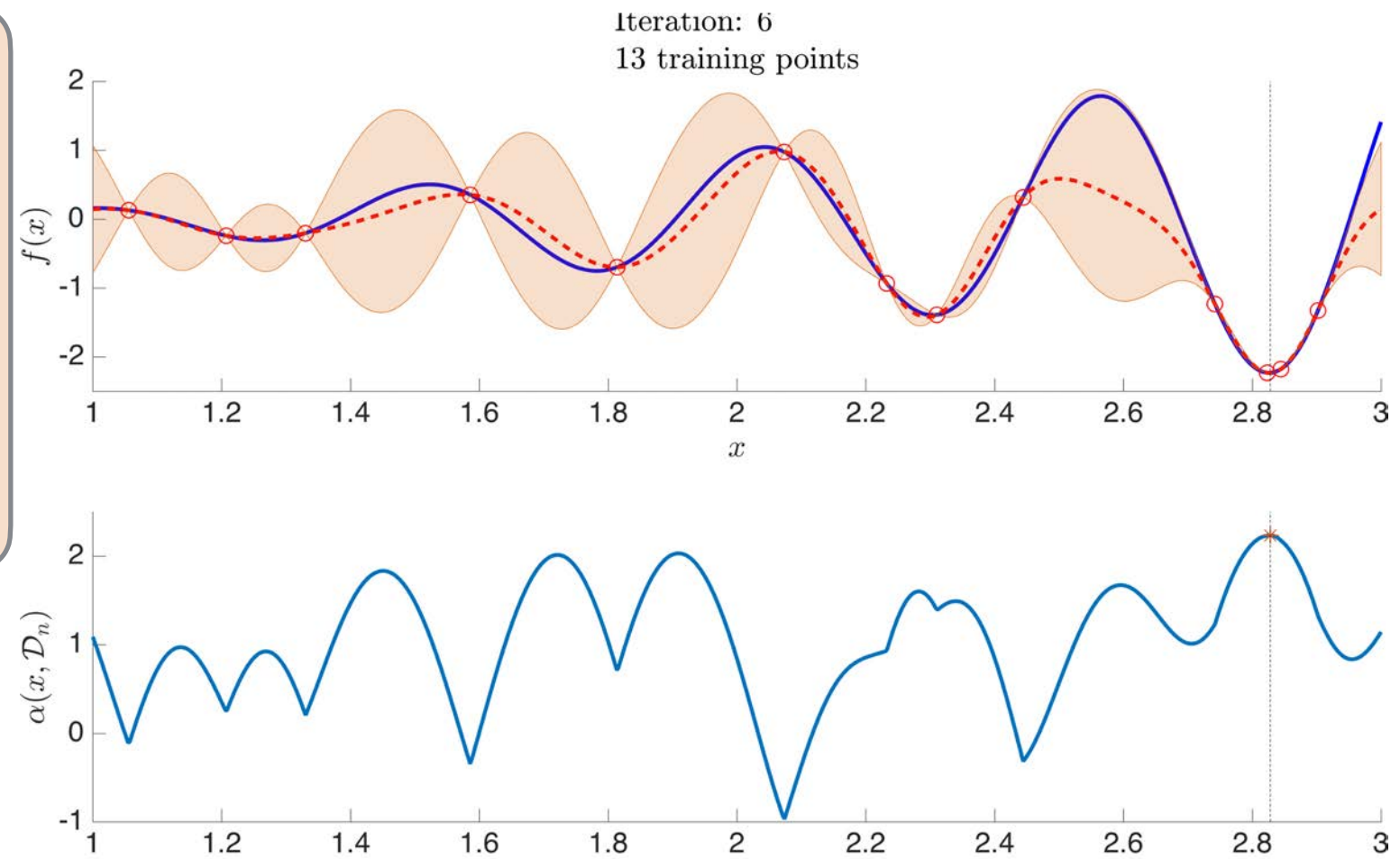
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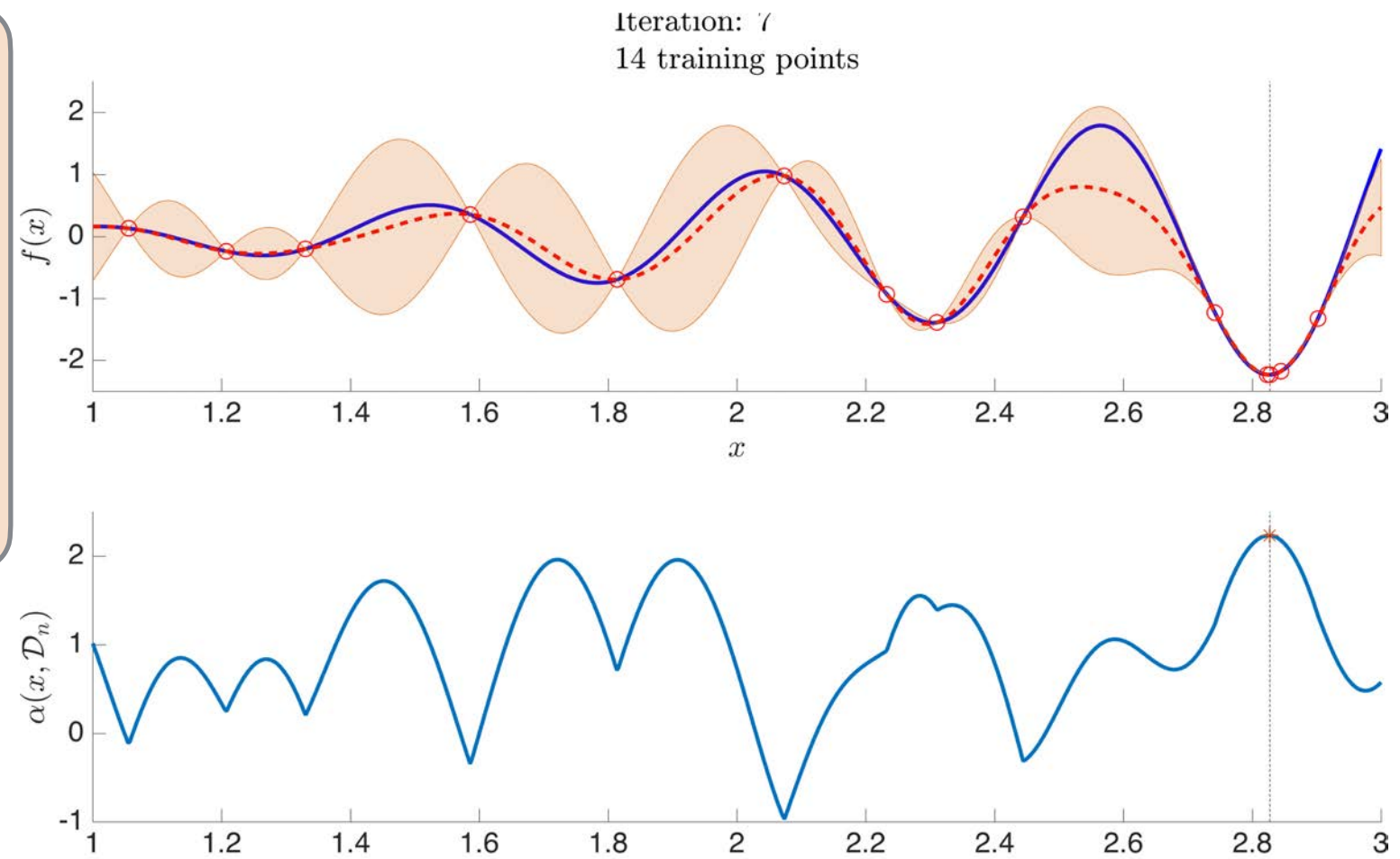
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Learning level sets

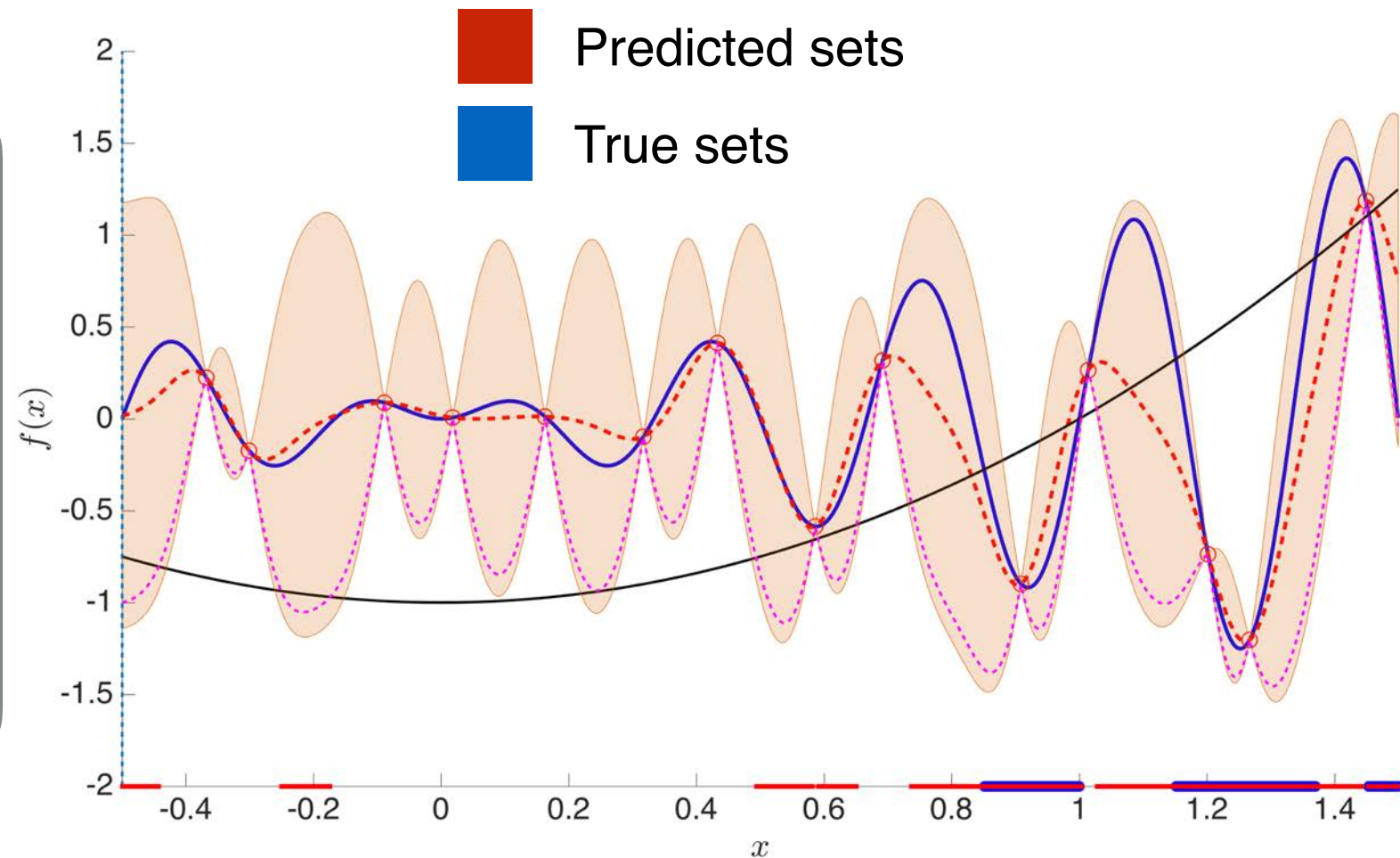
Goal: Identify the sets $L_\alpha(t) = \{x : \mathcal{R}_\alpha(f(x)) \leq t(x)\}$

Utilize the posterior to guide a sequential sampling policy by optimizing a chosen expected utility function

$$\alpha(\mathbf{x}; \mathcal{D}_n) = \mathbb{E}_\theta \mathbb{E}_{v|\mathbf{x}, \theta} [U(\mathbf{x}, v, \theta)]$$

e.g. sample at the locations that maximize the posterior variance in $L(t)$

$$\mathbf{x}_{n+1} = \arg \max_{\mathbf{x} \in L_\alpha(t)} V(f(\mathbf{x}))$$



Terminate iteration when the “volume” of the predicted level sets is below a given threshold:

$$|V_{n+1}(t) - V_n(t)| < \epsilon, \quad V_n(t) = \int_{L_\alpha(t)} \mathbf{1}_{[-\infty, t]} d\mathbf{x}$$

Remarks:

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Learning level sets

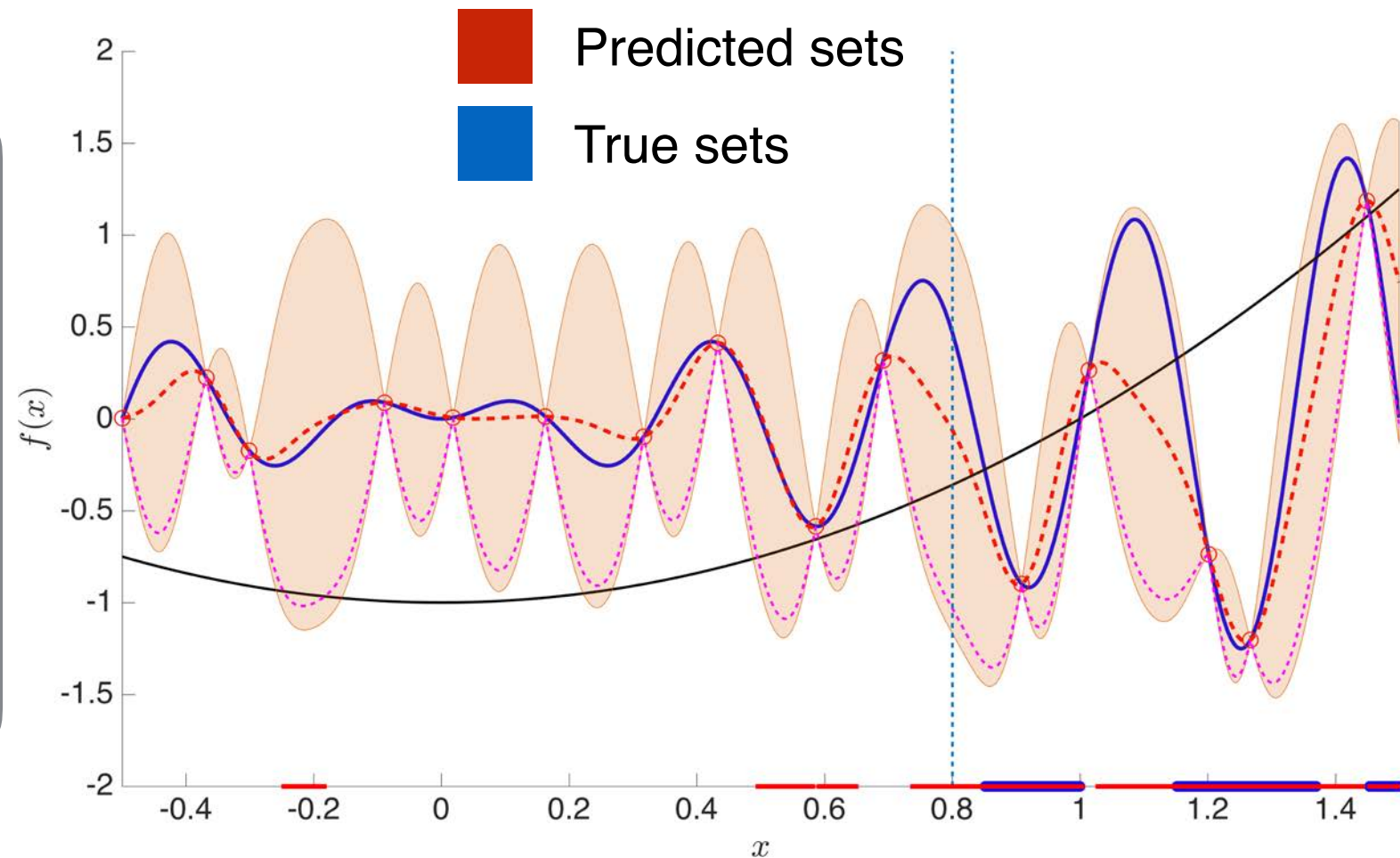
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Utilize the posterior to guide a sequential sampling policy by optimizing a chosen expected utility function

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e.g. sample at the locations that maximize the posterior variance in $L(t)$

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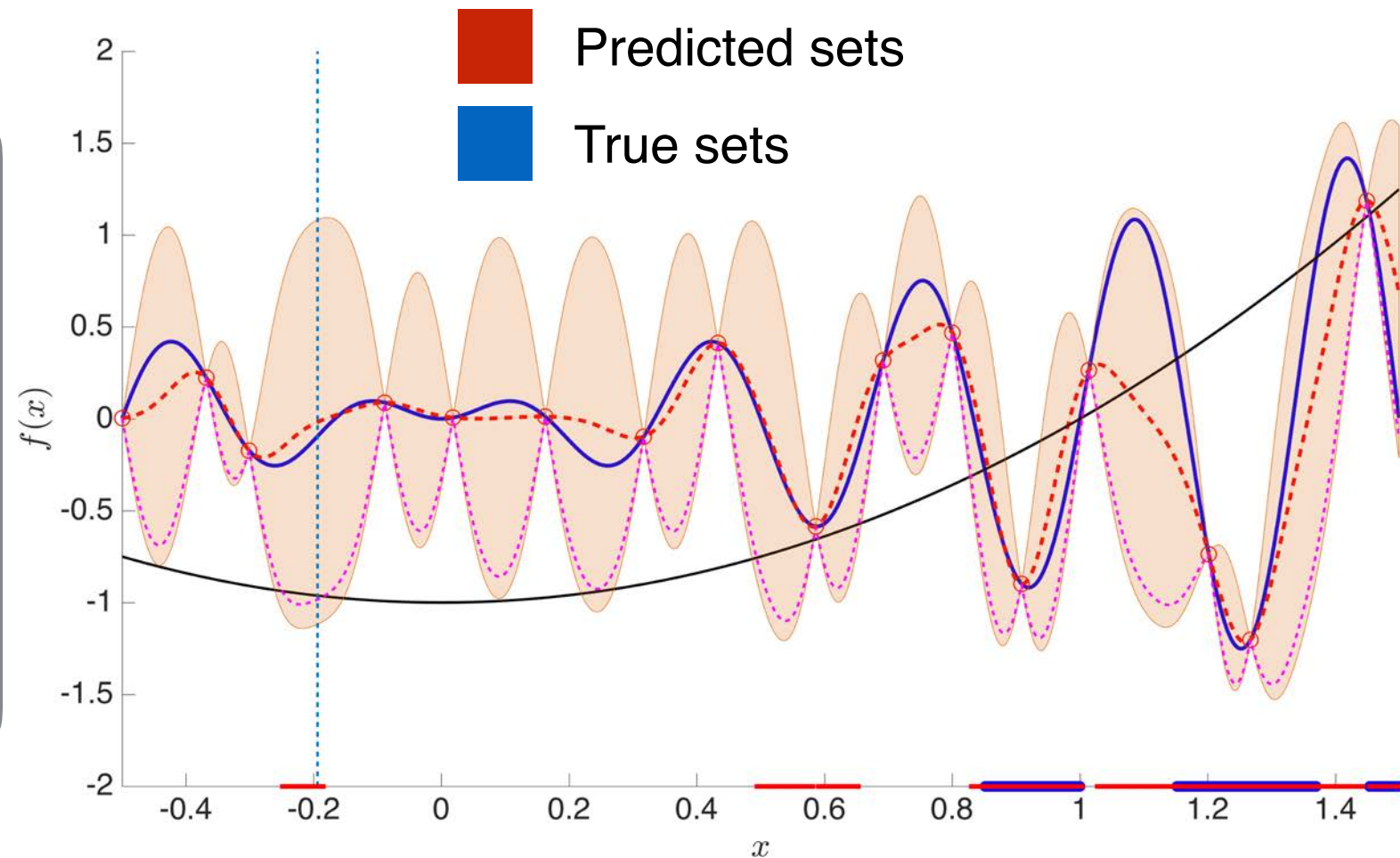
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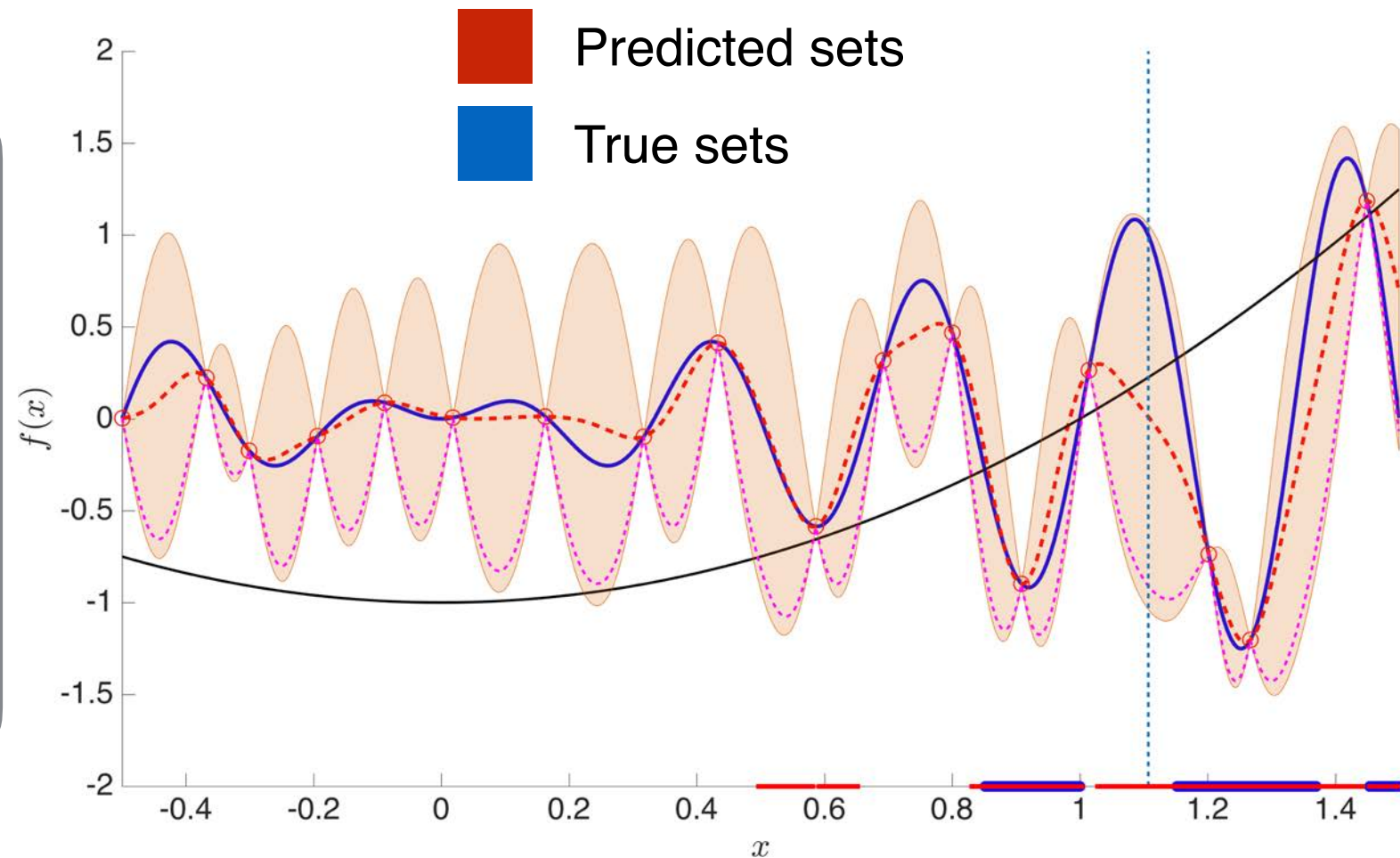
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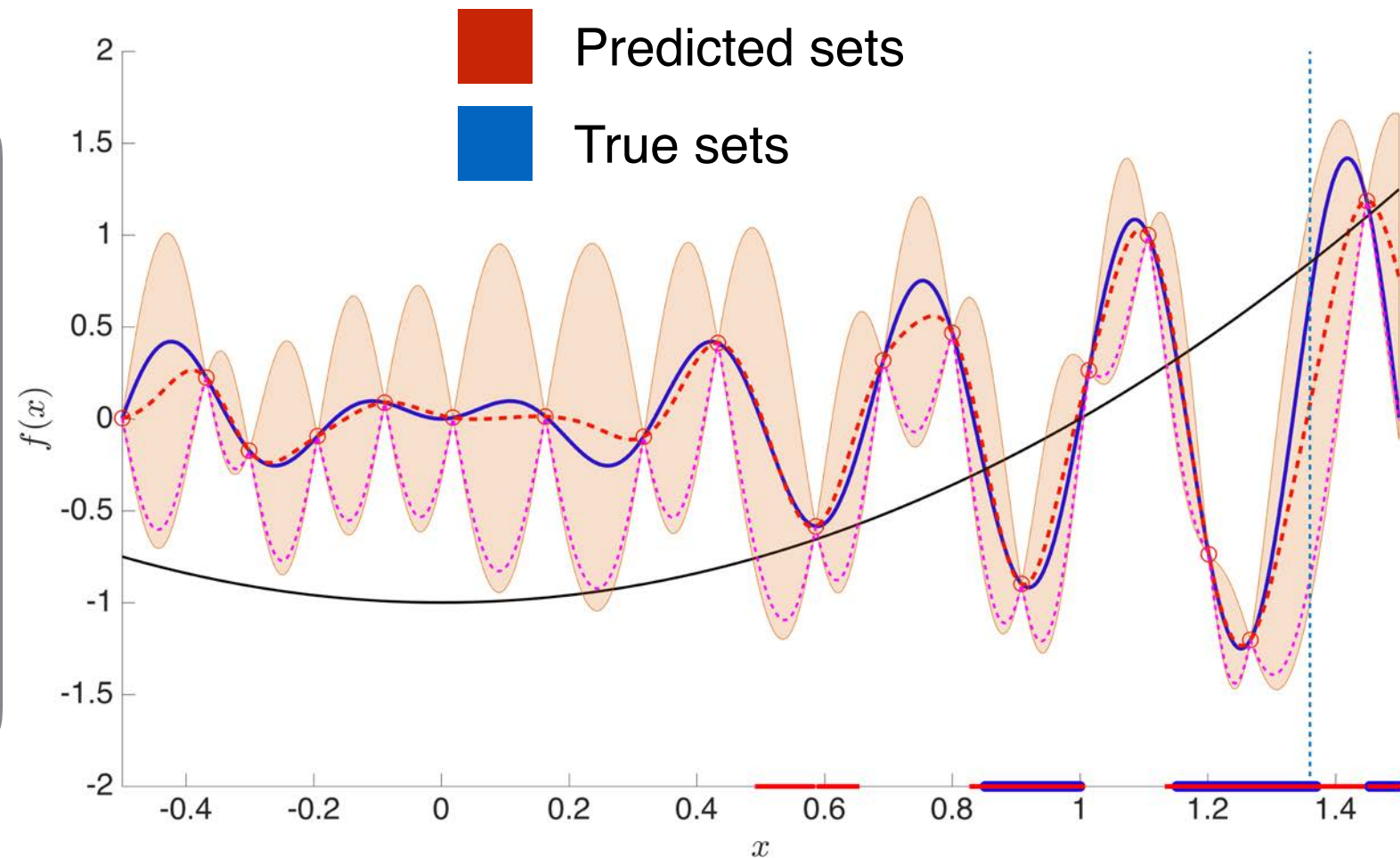
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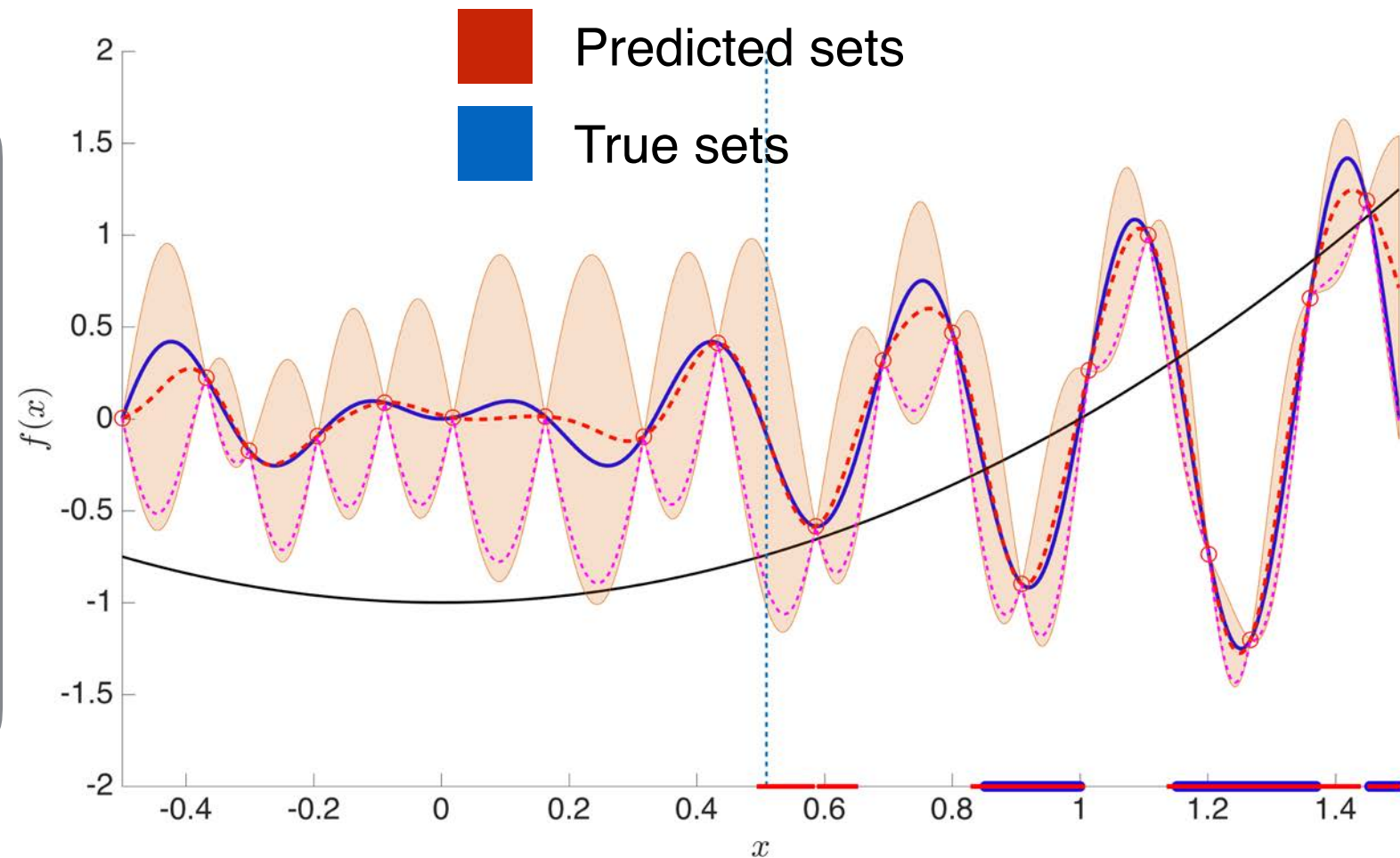
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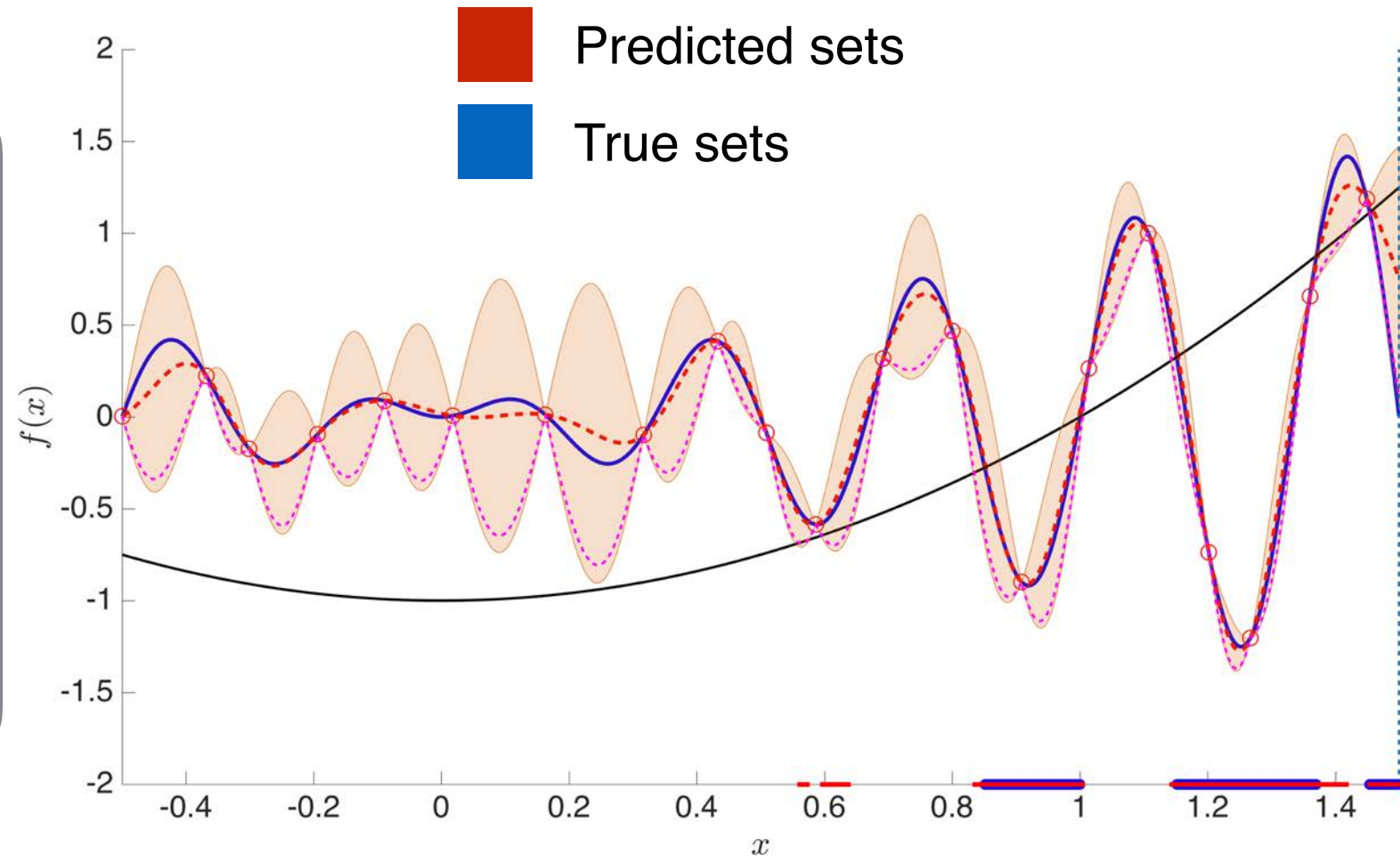
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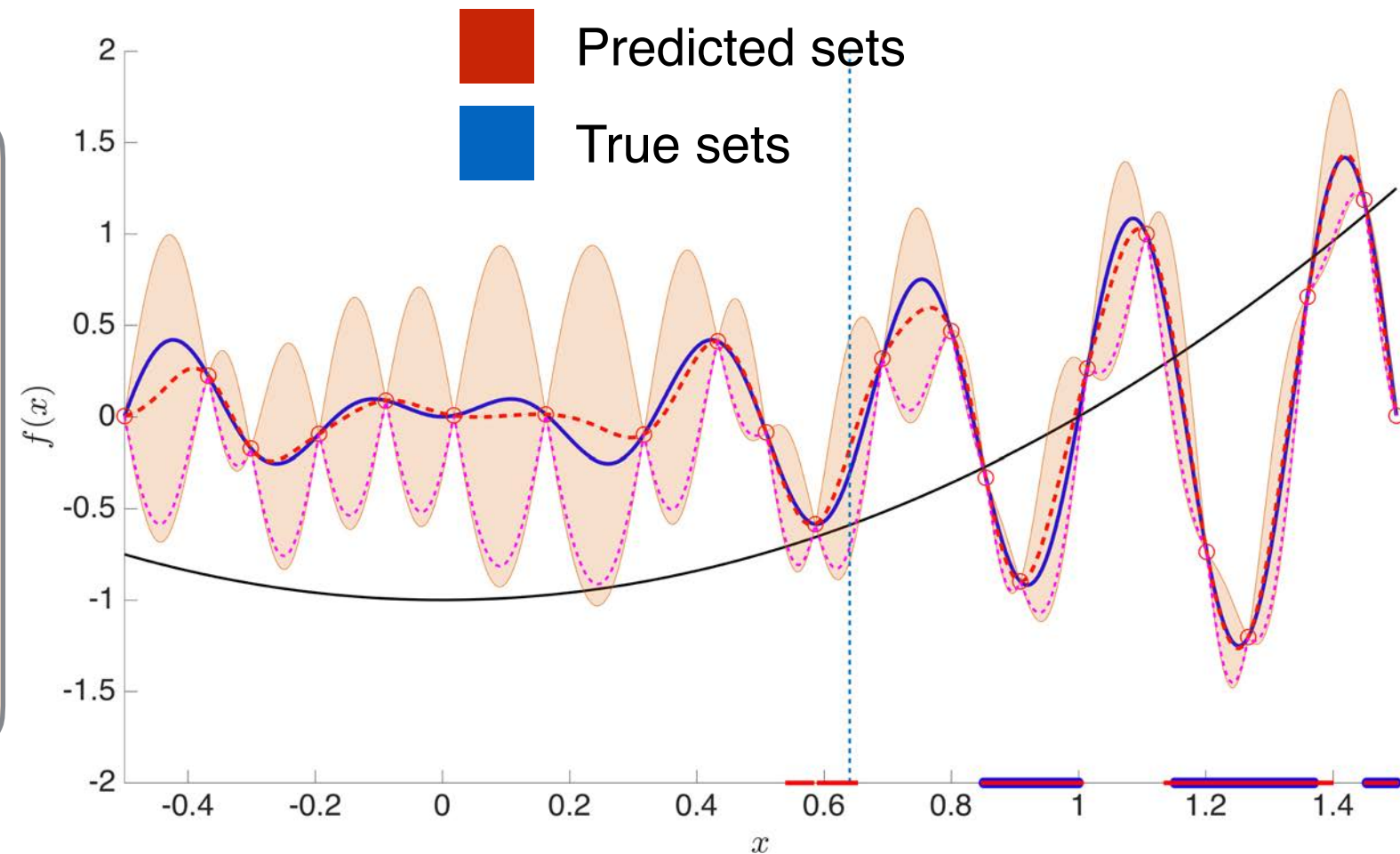
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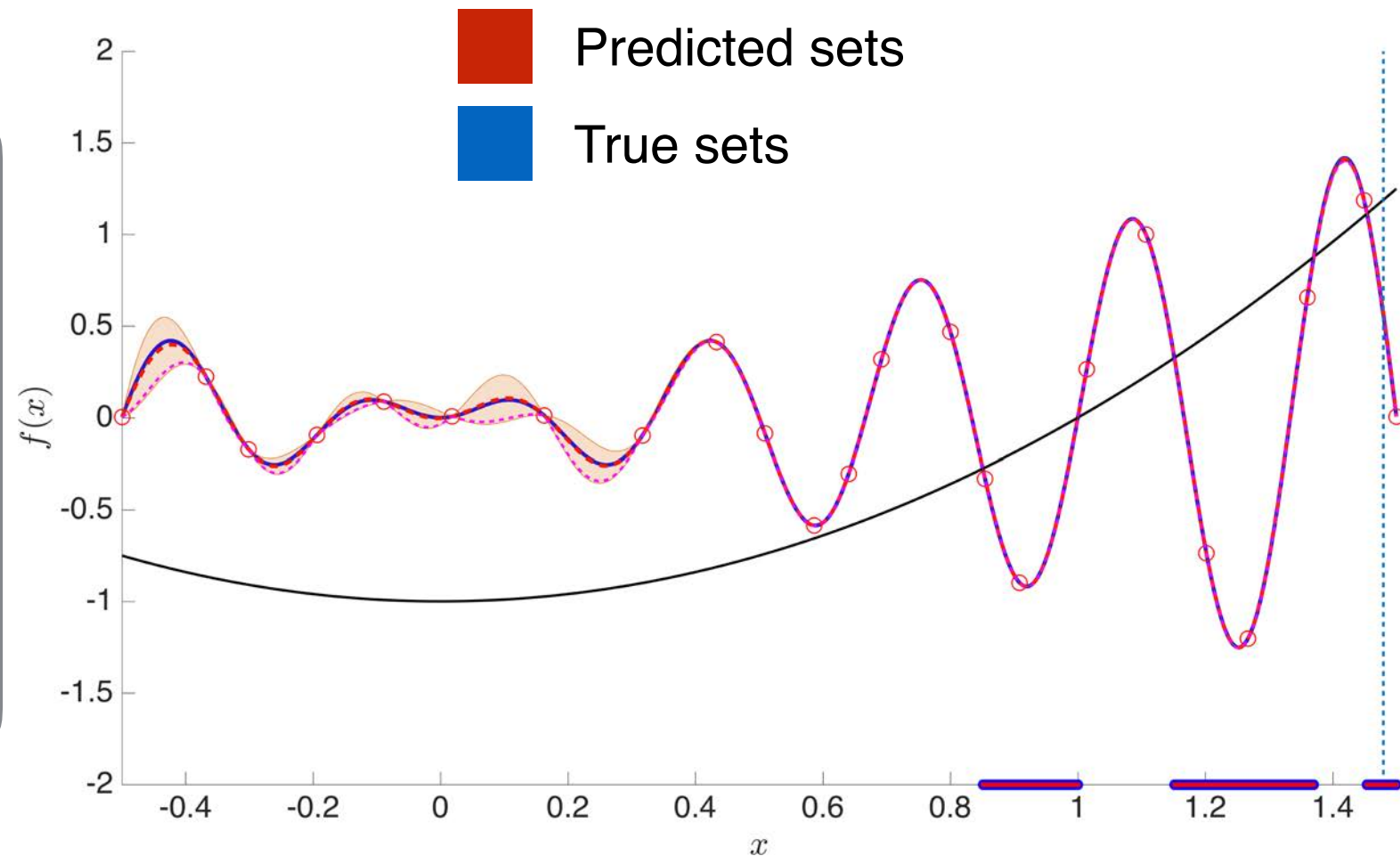
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